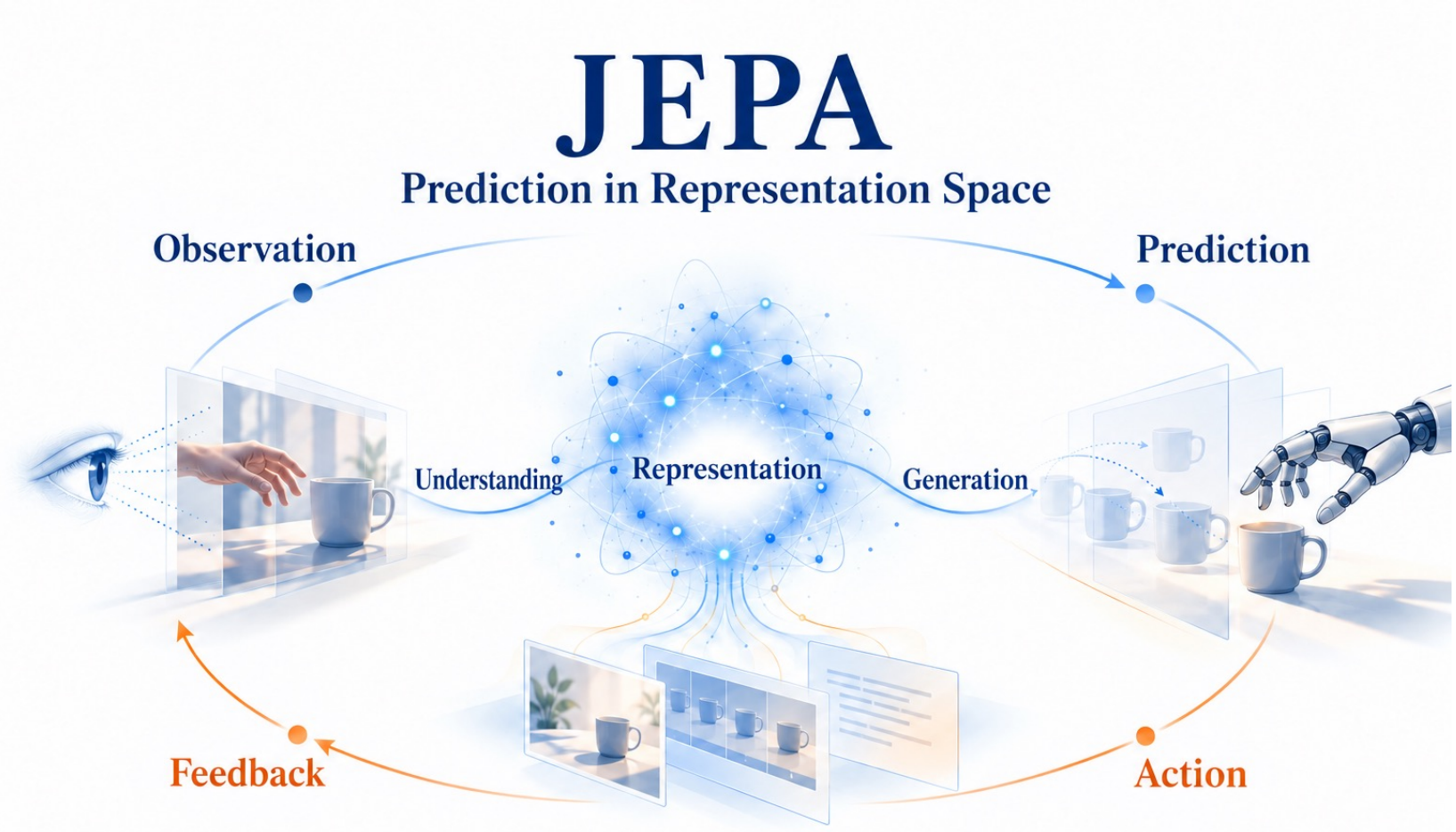
The JEPA Series: A Review

Tao Zhang · October 14, 2026



Outline

1. Motivation and representation learning
2. JEPA architectures
3. Preventing collapse
4. Applications across modalities
5. Synthesis and open questions

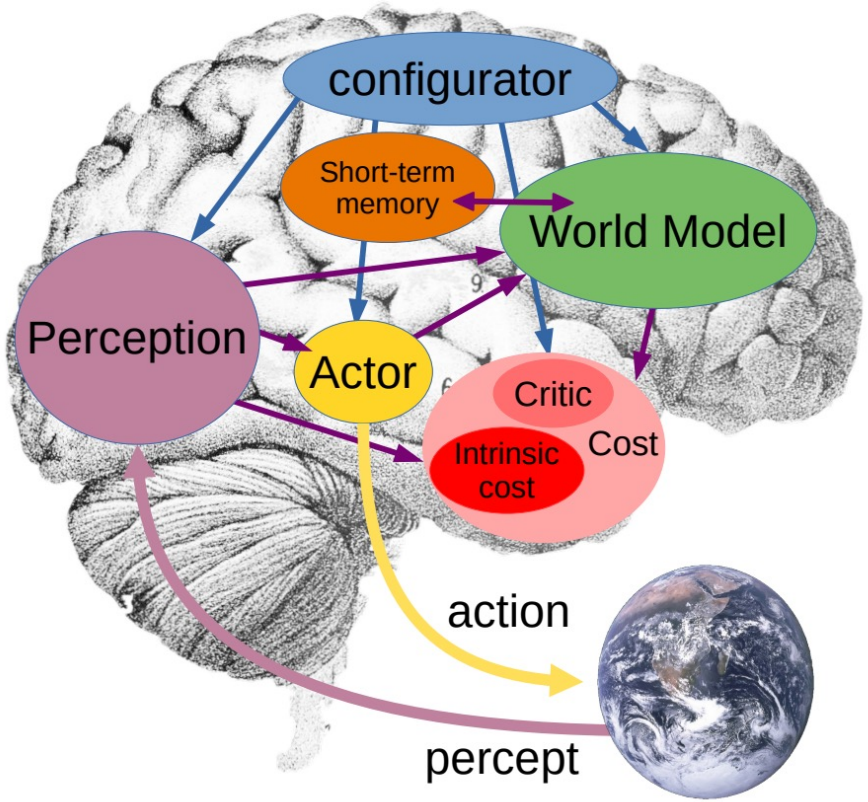
§1 Intelligence and world models

Prediction supports deliberation

Perception estimates the current state.

A world model predicts possible consequences.

An actor uses goals and costs to choose actions.



Perception, world model, cost and action

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.

§1 Representation, prediction and planning

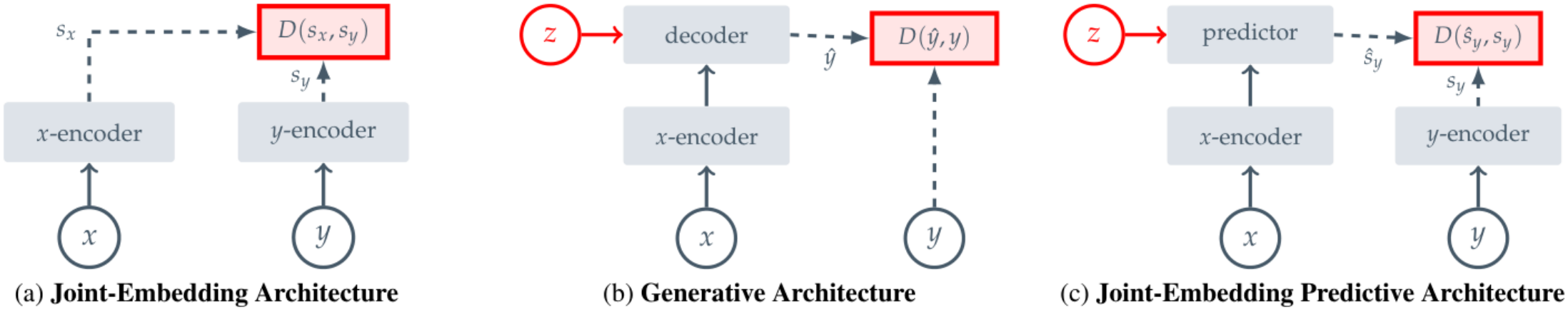
Each capability requires an additional component

Level	What it learns or does	What remains necessary
Representation	Predicts a feature target	Task-relevant information
World model	Predicts state changes	Actions and temporal structure
Planning	Searches for useful actions	Goals, costs and closed-loop execution
Autonomous system	Coordinates modules and tasks	Memory, configuration and reliability

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

§1 Prediction in representation space

The prediction target determines the learning problem



JEA, generative prediction and JEPA, I-JEPA Fig. 2

Generative prediction targets observations. JEPA targets learned features.

The encoder can discard detail, so the retained information matters.

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.

§1 JEPA design philosophy

Prediction defines what the representation must retain

Learn the target space

Predict learned features, without reconstructing every pixel.

Condition the prediction

Context and target position specify what to predict.

Control information loss

Avoid collapse and test whether useful details survive.

I-JEPA instantiates this idea with masked image-region prediction.

§1 I-JEPA components: networks and updates

The main architectural ingredients have clear precedents

Component	Role in I-JEPA	Earlier examples	Unique?
Context encoder	Encode visible context	MAE, CAE	No
Target encoder	Provide learned targets	BYOL, DINO, data2vec	No
EMA teacher	Smooth target updates	MoCo, BYOL, data2vec	No
Stop-gradient	Block target-branch gradients	BYOL, SimSiam, DINO	No
Predictor	Predict target features	BYOL, SimSiam, CAE	No
Visible-only encoding	Reduce encoder compute	MAE, CAE	No

Same component names can serve different prediction tasks.

I-JEPA: Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture · 2023-06 · CVPR
Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning · 2020-06 · arXiv preprint
Masked Autoencoders Are Scalable Vision Learners · 2021-11 · arXiv preprint

§1 I-JEPA components: targets and masking

The distinctive choices concern the target and prediction task

Design choice	Role in I-JEPA	Related work	Unique?
Masked prediction	Infer missing regions	MAE, BEiT, data2vec	No
Mask / position inputs	Specify target locations	MAE, BEiT, CAE	No
Feature-space targets	Predict learned representations	data2vec, CAE	No
Full-context teacher	Encode full image, then select targets	data2vec, iBOT	No
No negative pairs	Match targets without negatives	BYOL, DINO, data2vec	No
Multi-block masking	Large targets + informative context	I-JEPA mask ablations	Recipe*

*Distinctive recipe. Masking itself has prior work.

I-JEPA: Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture · 2023-06 · CVPR
data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language · 2022-07 · ICML
Context Autoencoder for Self-Supervised Representation Learning · 2022-02 · arXiv preprint

§1 Distinctive contributions and lessons

I-JEPA advances a task design and an effective training recipe

Task design

Large target blocks + informative context

Treat mask geometry as a main design variable.

Efficient recipe

Visible-only encoding + narrow predictor + EMA

Compare transfer at matched training compute.

Evidence and transfer

Useful frozen features without pixel reconstruction

For generation, also test reconstruction and detail retention.

I-JEPA: Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture · 2023-06 · CVPR
data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language · 2022-07 · ICML

§2 JEPA architectures

2

2.1 The original JEPA blueprint

2.2 Hierarchical prediction

2.3 Learning temporal change

§2 JEPA architectures: summary

The central choice is what becomes predictable

Work	Conditioning or structure	Main contribution	Boundary
JEPA, 2022	Feature predictor + latent z	Informative, predictable codes	Blueprint; no agent benchmark
H-JEPA concept	Multiple time scales	Coarse-to-fine planning	Proposal; learning unresolved
H-JEPA, 2026	Learned hierarchy + actions	Subgoal costs in learned spaces	Four simulators; DROID offline
TDV, 2026	Observed RGB difference	Explicit latent change	Dense-task trade-offs

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[8] H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning. October 2026. arXiv preprint.
[9] You Don't Need Strong Assumptions: Visual Representation Learning via Temporal Differences. June 2026. arXiv preprint.

§2 JEPA: the original blueprint

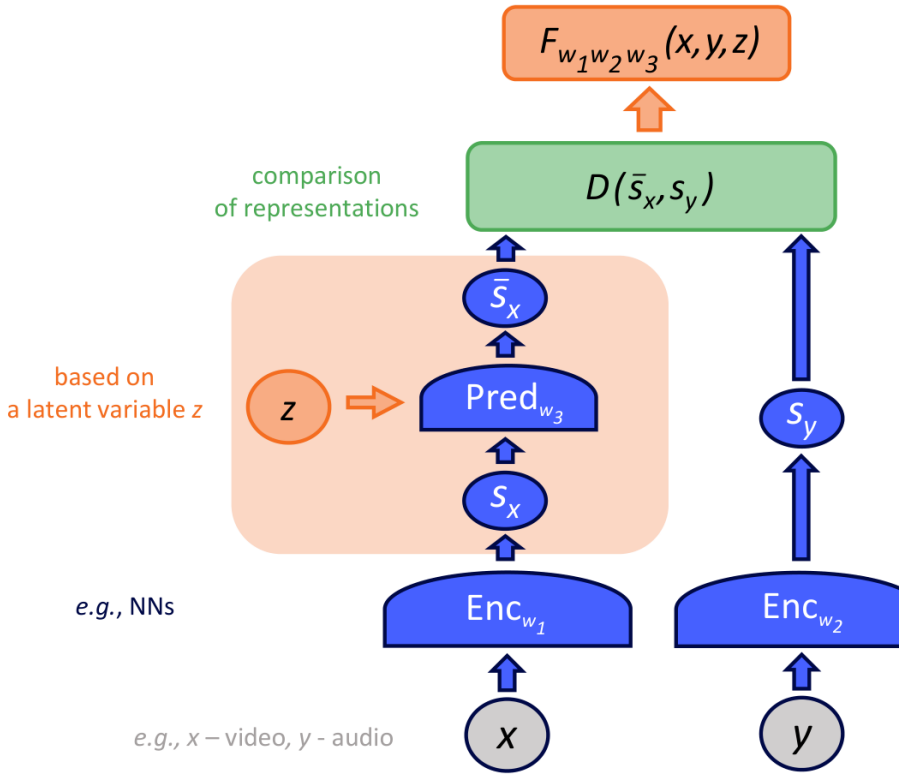
Predict one representation from another

Encode paired observations into features.
Predict target features from context and conditioning.

$$E(x, y, z) = D\big(p(s_x, z), s_y\big)$$

s_x,s_y: features of x,y. p: predictor.
z: condition or latent. D: feature mismatch.
E: scalar energy.

Training must preserve information.



JEPA architecture, Dawid and LeCun Fig. 9

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.

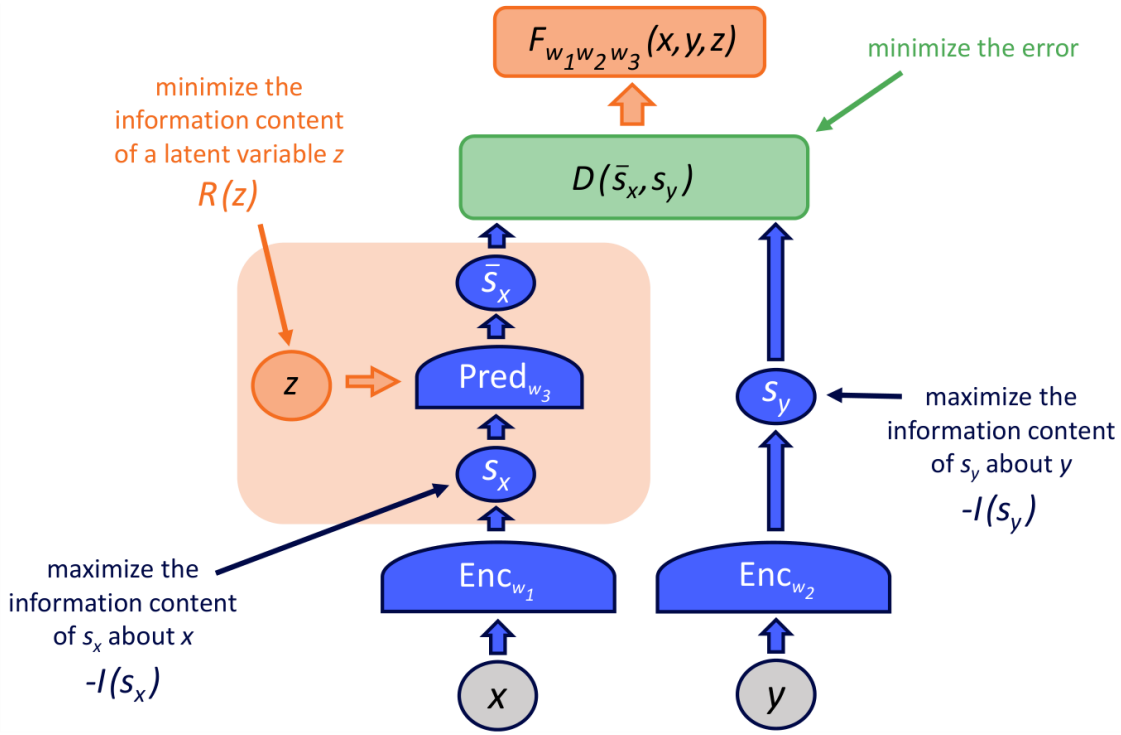
§2 The conditioning variable z

Uncertainty needs a controlled information path

z supplies information missing from context.
Limit its capacity so it cannot copy the target.

$$F(x, y) = \min_{z}[E(x, y, z) + R(z)]$$

z: free latent. R: capacity penalty.
E: prediction mismatch. F: minimum energy.
Observed actions / positions remain fixed.



Prediction and information constraints, Fig. 10

Figure: s_x,s_Y are features; I(·) denotes retained information.

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.
[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

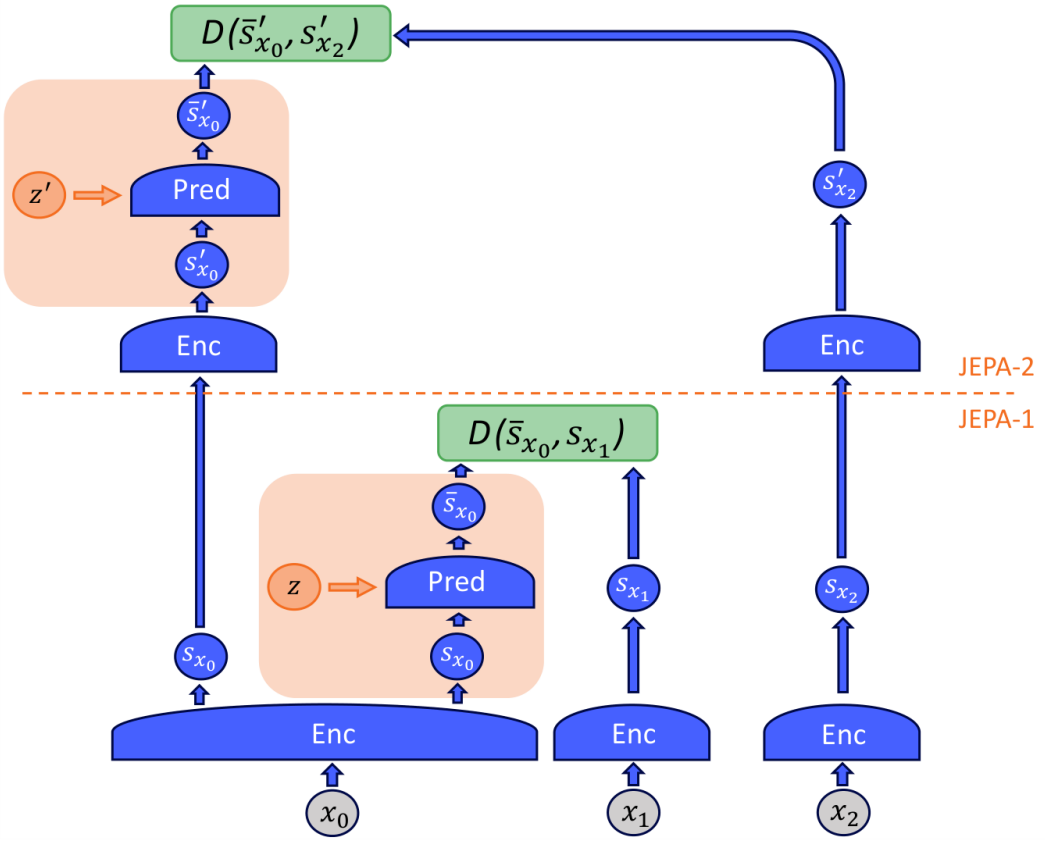
§2 H-JEPA: the hierarchy concept

Abstraction changes with the prediction horizon

Lower levels retain detail for short horizons.

Higher levels predict coarser states over longer horizons.

Hierarchical planning also needs compatible subgoals.



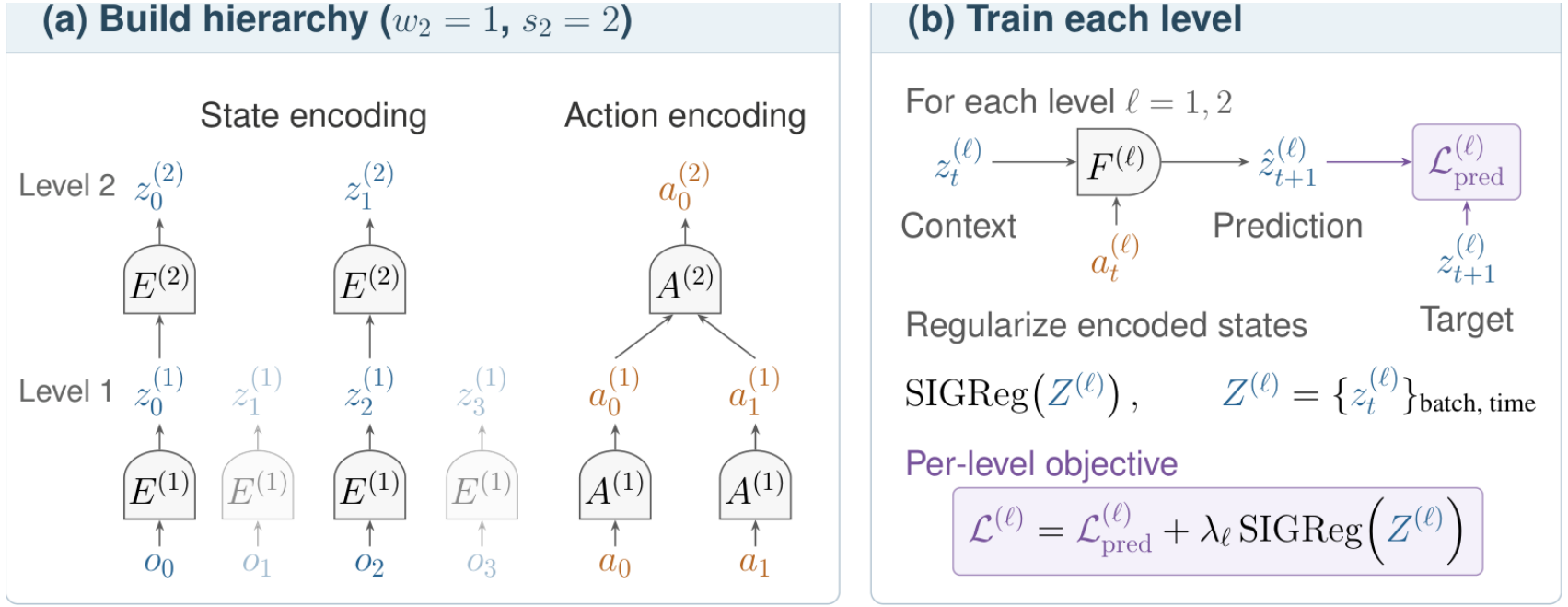
Two levels of predictive abstraction, Fig. 11

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.

[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.

§2 H-JEPA: visual planning

Learn separate spaces at different time scales



Hierarchical training, H-JEPA Fig. 2

Level-specific dynamics use increasing temporal strides.

Top-down subgoals improve long-horizon planning in the reported environments.

[8] H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning. October 2026. arXiv preprint.

§2 TDV: learning temporal change

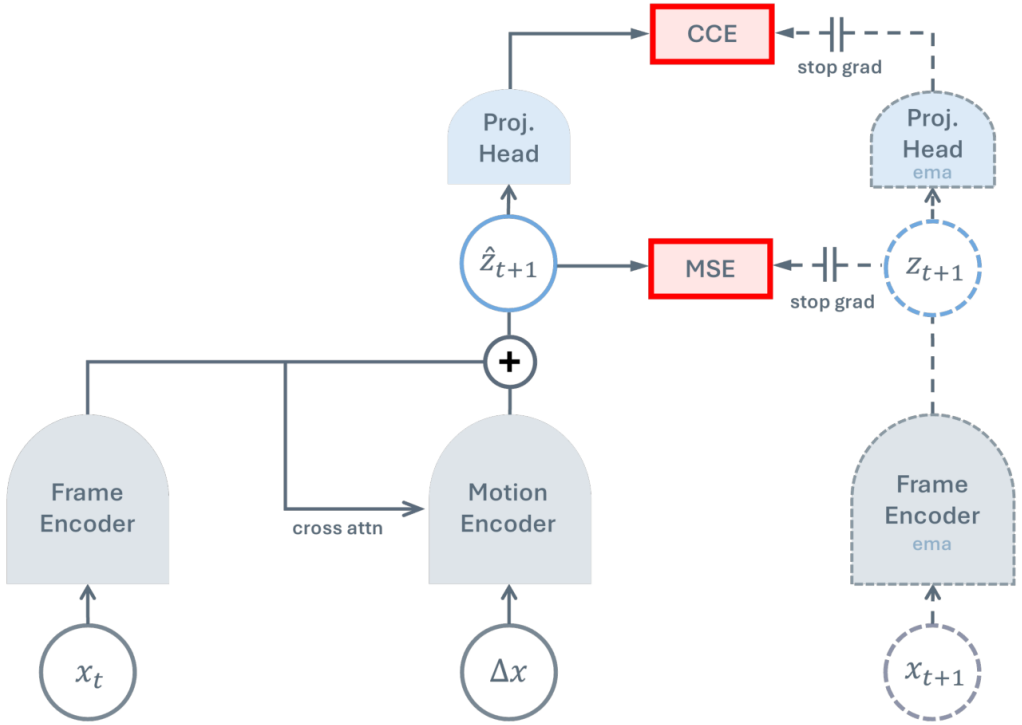
Represent change instead of enforcing invariance

Learn change between adjacent frame representations.

$$\Delta z_t = m_\phi(\Delta x_t; z_t)$$

$$\hat{z}_{t+1} = z_t + \Delta z_t$$

$\Delta x_t = x_{t_{+1}} - x_t$: observed RGB change.
z_t / Δz_t: features / predicted change.
m_ϕ: motion encoder. $\hat{z}_{t_{+1}}$: predicted features.



Frame and motion encoders, TDV Fig. 2

Both frames are observed; training uses EMA self-distillation.

[9] You Don't Need Strong Assumptions: Visual Representation Learning via Temporal Differences. June 2026. arXiv preprint.

§2 JEPA architectures: summary

The central choice is what becomes predictable

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JEPA, 2022	Feature predictor + latent z	Informative, predictable codes	Blueprint; no agent benchmark
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[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[8] H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning. October 2026. arXiv preprint.
[9] You Don't Need Strong Assumptions: Visual Representation Learning via Temporal Differences. June 2026. arXiv preprint.

§3 Preventing collapse

3

3.1 Why matching features can collapse

↓

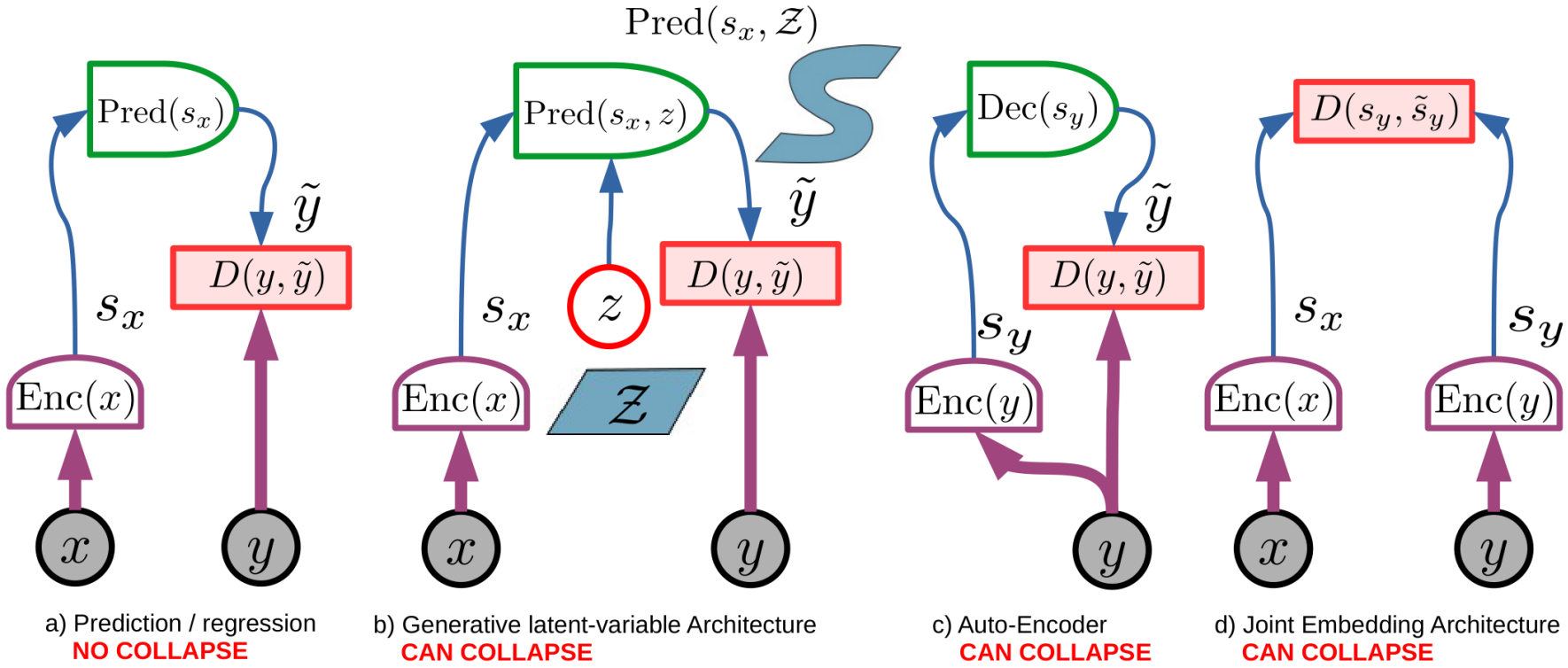
3.2 Contrast, asymmetry and regularization

↓

3.3 Scale, shape and useful information

§3 Why low energy can be uninformative

Energy collapse: incompatible inputs also receive low energy.



Energy-based architectures and collapse · LeCun, 2022, Fig. 10

Fitting observed pairs is insufficient without constraining other pairs.

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.

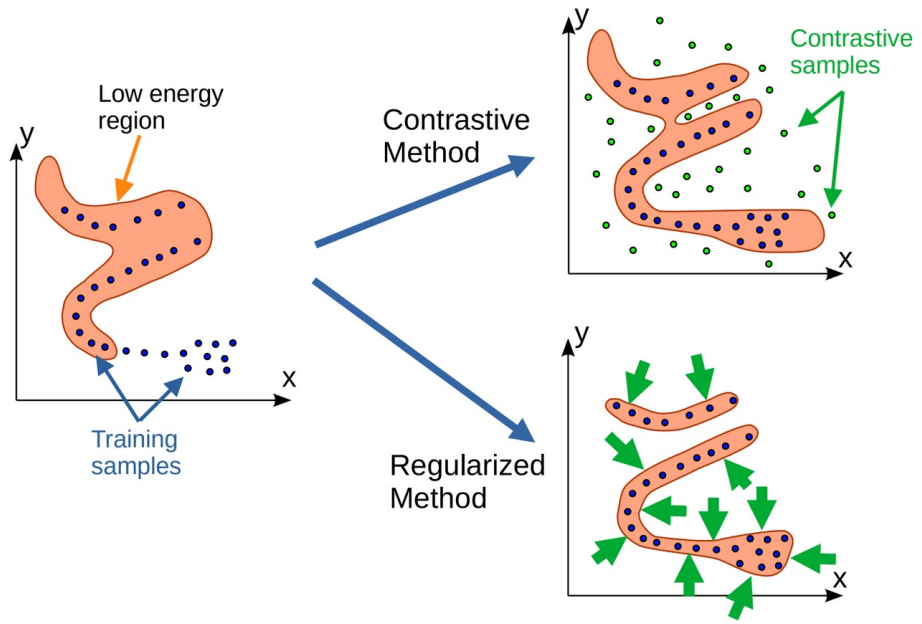
§3 Why feature prediction can collapse

A constant target can make prediction trivial

$$f(x) = c, \quad p(c) = c \quad \Rightarrow \quad \mathcal{L}_{\text{pred}} = 0$$

x: input
f: encoder
p: predictor
c: the same code for every input

L_{pred}: feature-matching loss.
Matching alone permits this solution.



Contrastive and regularized approaches

[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.
[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.
[11] Bootstrap Your Own Latent - A New Approach to Self-Supervised Learning. December 2020. NeurIPS.

§3 Collapse and information loss

Healthy statistics can still miss structure and action information

Failure	What fails	Control	Examples	Check
Constant code	Same vector for all inputs	Separation or asymmetry	MoCo / BYOL / SimSiam / I-/V-JEPA	Batch spread
Low spread / redundant axes	Low variance / few axes	Std + covariance	VICReg	Std + spectrum
Output collapse	Same probabilities	Center + sharpen	DINO	Output diversity
Joint dependence	Hidden linear dependencies of different dimmension	Projected laws	LeJEPA (SIGReg) VISReg (SWD)	Projections
Action blindness	Lost action effects	Transition losses	AVL-JEPA	Action probes

Joint dependence and action blindness extend beyond classical collapse.

[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

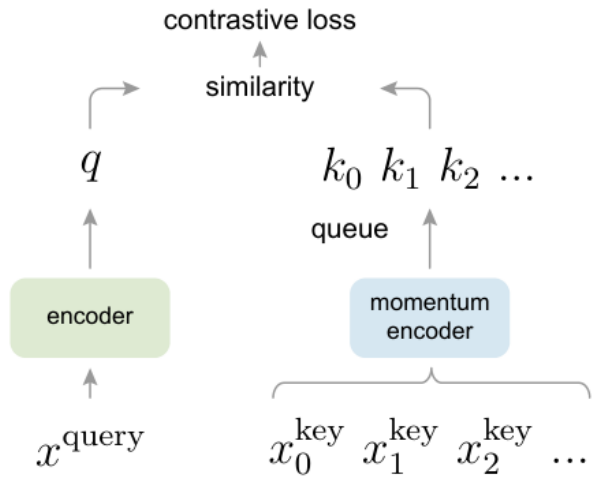
§3 MoCo: contrastive learning

The paired key competes with a queue of other keys

$$\ell_i = -\log \frac{\exp\left(q_i^\top k_i^+ / \tau\right)}{\exp\left(q_i^\top k_i^+ / \tau\right) + \sum_{j=1}^{K} \exp\left(q_i^\top k_j^- / \tau\right)}$$

q_i: query for image i
k_i^+: its paired key
k_j^-: a cached negative key

All q and k vectors have unit L2 norm.
K: queue size. $\tau > 0$: temperature.

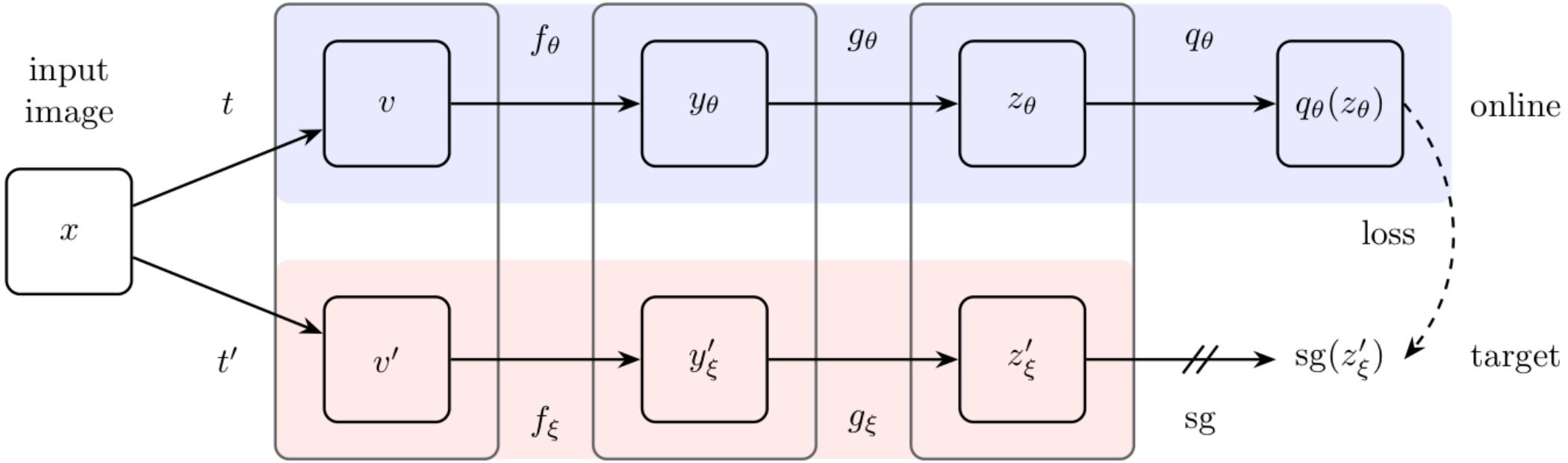


Momentum keys and queue, MoCo Fig. 1

[12] Momentum Contrast for Unsupervised Visual Representation Learning. June 2020. CVPR.

§3 BYOL and SimSiam: asymmetric learning

Asymmetric training supports learning without negatives



Online and target branches in BYOL, Fig. 2

BYOL: predictor + stopped EMA target. SimSiam: predictor + stop-gradient with shared weights.

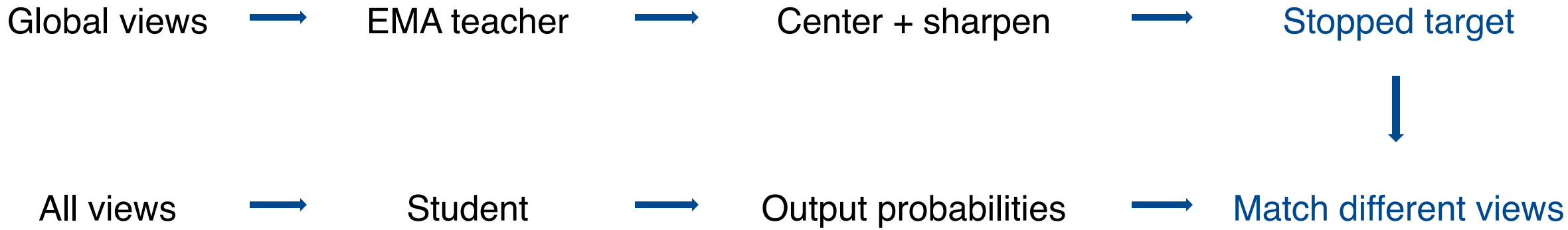
f/g/q: encoder/projector/predictor. θ/ξ: online/teacher; t/t′: views. sg: stop-gradient.

[11] Bootstrap Your Own Latent - A New Approach to Self-Supervised Learning. December 2020. NeurIPS.

[13] Exploring Simple Siamese Representation Learning. June 2021. CVPR.

§3 DINO: control the output distribution

Center and sharpen the teacher target, then match views



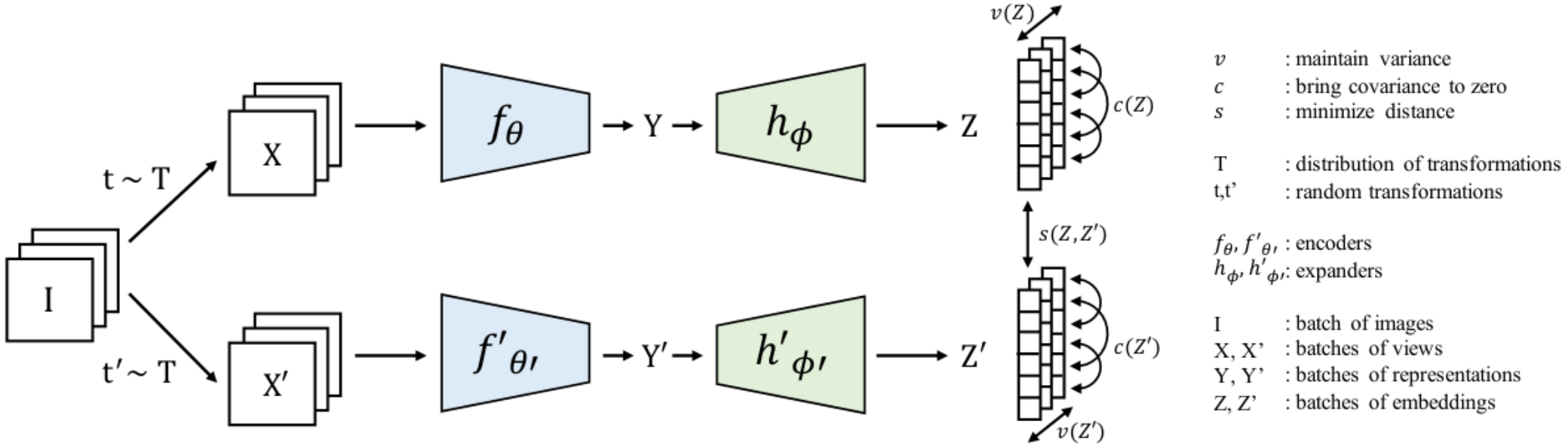
Centering limits dominant outputs; sharpening avoids uniform targets.

EMA: the teacher follows a moving average of student weights.

[7] Emerging Properties in Self-Supervised Vision Transformers. October 2021. ICCV.

§3 VICReg: explicit non-collapse constraints

Separate agreement, activity and redundancy



The three VICReg terms, Fig. 1

Paired views agree, each dimension stays active, and off-diagonal covariance shrinks.

No negative pairs or momentum teacher are required by the recipe.

[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.

§3 VICReg: three loss terms

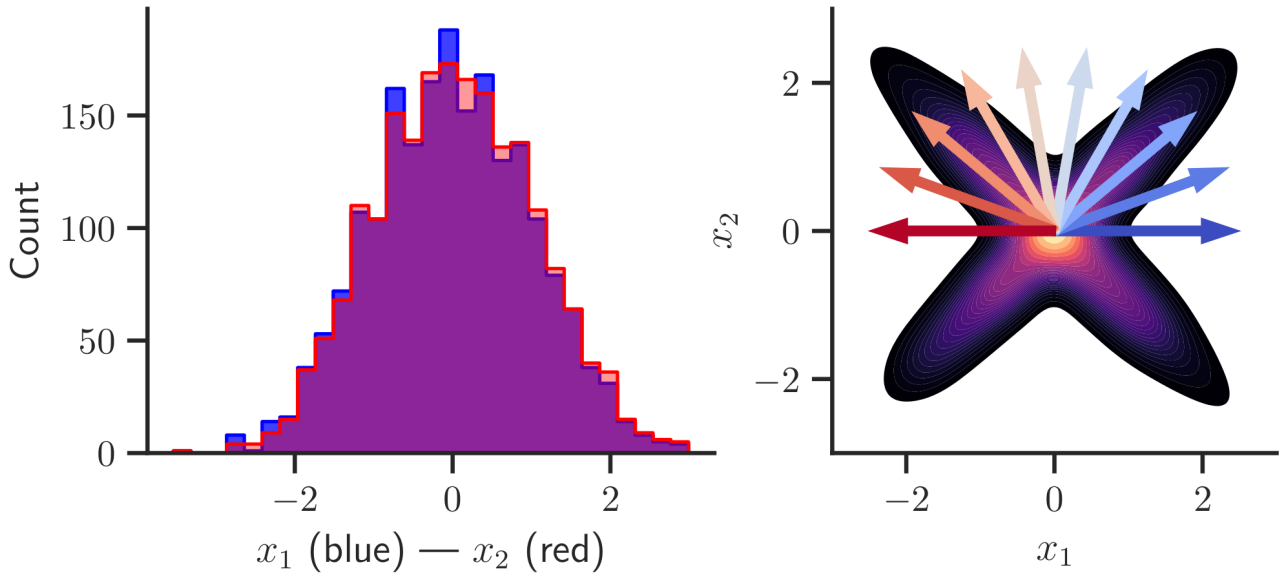
Each term controls a different part of the representation

Term	Definition	Role
Invariance	$s(Z, Z') = \frac{1}{ND}\sum_{i=1}^{N}\sum_{j=1}^{D}\left(Z_{ij} - Z'_{ij}\right)^2$	Match paired views
Variance	$v(Z) = \frac{1}{D}\sum_{j=1}^{D}\left[1 - \sigma_j(Z)\right]_+$	Protect coordinate spread
Covariance	$c(Z) = \frac{1}{D}\sum_{j\neq k} C_{jk}\,(Z)^2$	Reduce linear redundancy

[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.

§3 VICReg: limits of second-order statistics

Gaussian marginals and zero covariance can hide dependence



Coordinate marginals and joint structure, LeJEPA Fig. 5

An exact counterexample

$$U \sim \mathcal{N}(0,1), \quad X = (U, SU)$$

S is an independent, equally likely +1 / –1 sign.

$$X_1, X_2 \sim \mathcal{N}(0,1), \quad \mathrm{Cov}(X) = I_2$$

$$Y = \frac{X_1 + X_2}{\sqrt{2}}, \quad \Pr(Y = 0) = \frac{1}{2}$$

I_2: 2 × 2 identity. Y: 45° projection.

The diagonal projection has a point mass. It is not Gaussian.

[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.

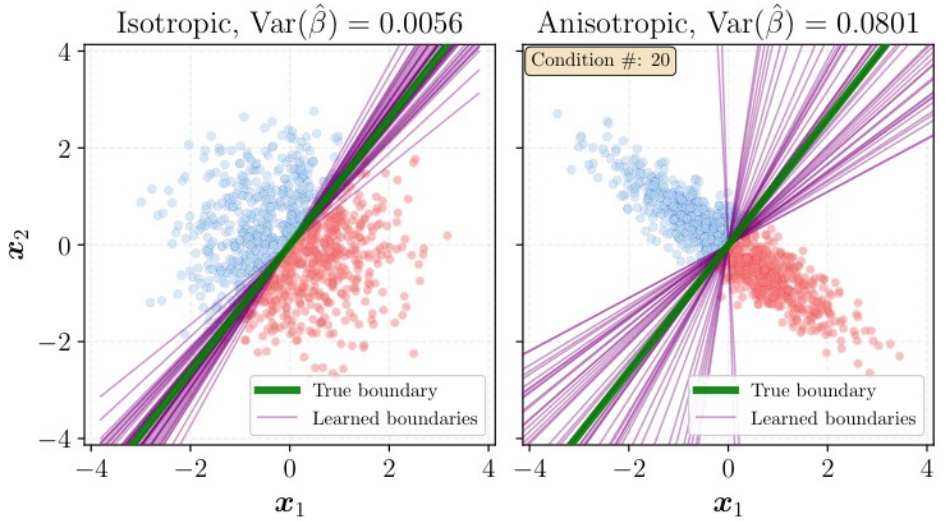
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

§3 LeJEPA: a distributional target

View agreement and distribution shape have different jobs

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_{\text{pred}} + \lambda R_{\text{SIG}}$$

L_{pred}: mean-squared distance to each image's global-view mean.
R_{SIG}: Gaussian-matching loss across samples, averaged over views.
$\lambda \in [0,1]$: the regularization weight.



Isotropy and probe variability, LeJEPA Fig. 3

The Gaussian optimum depends on the paper's feature and probe assumptions.

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

§3 LeJEPA: agree across views

Each image's global-view mean is a shared target

N images; V views each; first G views are global. D: embedding dimensions.
f_θ: shared encoder + projector; x_{nv}: view v of image n.

$$z_{nv} = f_\theta(x_{nv}) \in \mathbb{R}^D, \qquad c_n = \frac{1}{G}\sum_{g=1}^{G} z_{ng}$$

$$\mathcal{L}_{\text{pred}} = \frac{1}{NVD}\sum_{n=1}^{N}\sum_{v=1}^{V} \| z_{nv} - c_n \|_2^2$$

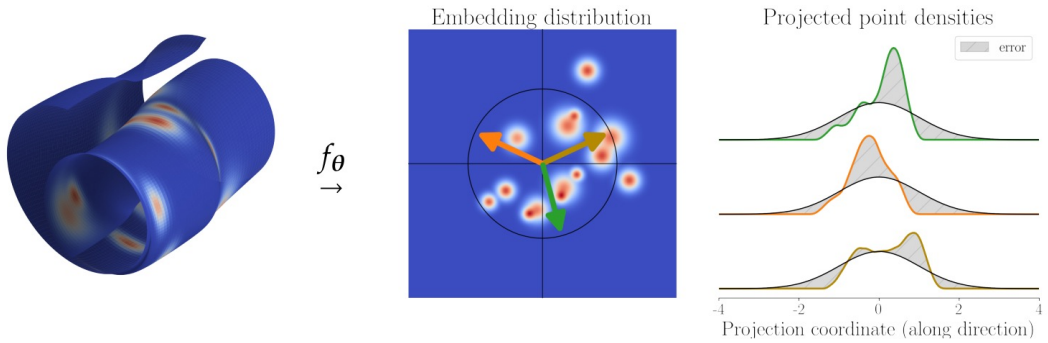
z_{nv}: embedding. c_n: its image's global-view mean. L_{pred}: view-agreement loss.

Add SIGReg across samples; gradients also pass through the target mean.

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

§3 SIGReg: Gaussianity through projections

From second moments to entire projected distributions



Random directions and Gaussian targets, LeJEPA Fig. 2

Z: N × D embeddings; z_n: sample n.
a_k: random unit direction, k = 1,…,K.
p_{nk}: scalar projection; t: frequency; $i^2 = -1$.
ϕ_k: empirical characteristic function.
$\exp(-t^2/2)$: Gaussian target and frequency weight.
Finite slices and numerical integration approximate the match.

$$p_{nk} = z_n^\top a_k, \qquad \hat{\varphi}_k(t) = \frac{1}{N}\sum_{n=1}^{N} e^{itp_{nk}}$$

$$R_{\mathrm{SIG}}(Z) = \frac{N}{K}\sum_{k=1}^{K}\int_{\mathbb{R}} \left|\hat{\varphi}_k(t) - e^{-t^2/2}\right|^2 e^{-t^2/2}\, dt$$

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

§3 VISReg: gradients under controlled rescaling

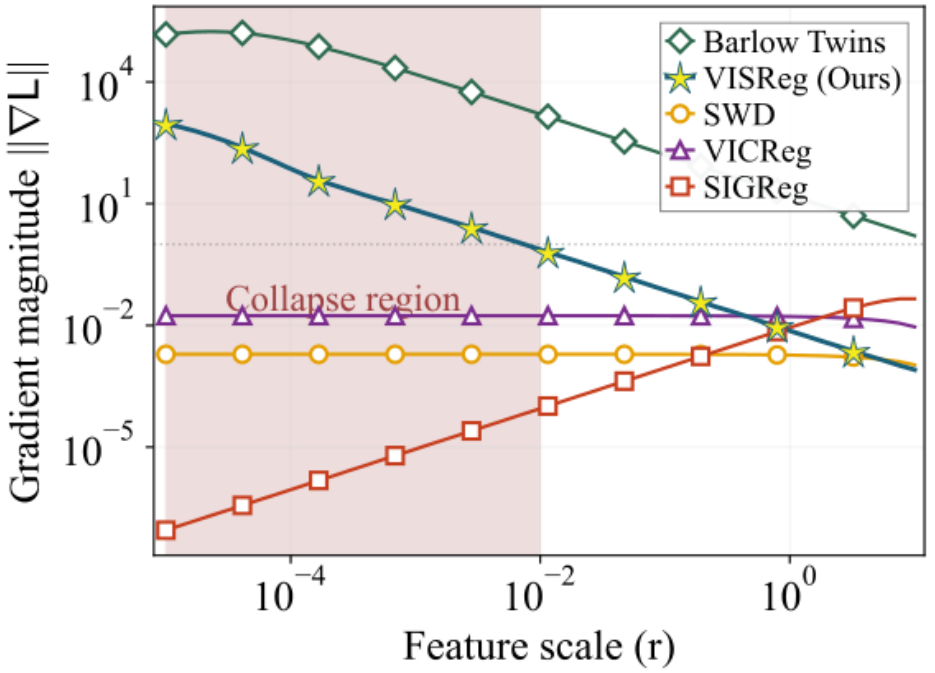
The figure probes gradients as feature scale decreases

Rescaling illustration

$$Z_r = rZ_0, \qquad r > 0$$

Z_0: fixed feature batch
r: imposed feature scale
Z_r: rescaled features

SIGReg gradients weaken near collapse in this diagnostic.



Gradient norm versus feature scale, VISReg Fig. 2

This is not a theorem about recovery during full training.

[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.

§3 VISReg: separate scale from shape

Separate scale recovery from shape adjustment

Scale recovery

Raw Z → Std σ → Unit spread

$$L_{\text{scale}} = \frac{1}{D}\sum_{j=1}^{D}\left(1-\sigma_j\right)^2$$

Shape adjustment

Center → Normalize → SWD

$$(Z_s)_{nj} = \frac{Z_{nj}-\mu_j}{\text{sg}(\sigma_j)+\epsilon}$$

Z: N samples × D coordinates; n: sample; j: coordinate.
μ_j, σ_j: coordinate mean and population std (N denominator).
Z_s: normalized features. sg: stop-gradient through the divisor only; $\varepsilon > 0$: division guard.

Scale has its own loss; SWD adjusts the normalized shape.

[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.

§3 SWD: match sorted projected samples

Adjust normalized shape by matching Gaussian quantiles

Normalize → Project → Sort samples → Match quantiles

$$p_{(i)k} = \mathrm{sort}(Z_s a_k)_i, \qquad q_i = \Phi^{-1}\left(\frac{i}{N+1}\right)$$

$$L_{\mathrm{shape}} = \frac{1}{NK}\sum_{k=1}^{K}\sum_{i=1}^{N}\left(p_{(i)k} - q_i\right)^2$$

Z_s: features centered and divided by detached coordinate std (+ε), ε > 0.
N samples; K random unit directions a_k. i: sorted rank; $p_{(i)k}$: sorted projection.
q_i: fixed target; Φ^{-1}: standard-normal quantile function.
Finite slices and quantile grids approximate distribution matching.

[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.

§3 VISReg: three controls, separate weights

Tune scale recovery and shape adjustment separately

$$R_{\mathrm{VIS}}(Z) = \alpha_s L_{\mathrm{scale}}(Z) + \alpha_h L_{\mathrm{shape}}(Z) + \alpha_c L_{\mathrm{center}}(Z)$$

L_{scale} Scale recovery: mean squared deviation of std from one.

L_{shape} Shape adjustment: mean squared gaps from sorted projections to Gaussian quantiles.

L_{center} Centering: mean squared coordinate means.

Z: embedding batch. α_s, α_h, $\alpha_c \geq 0$: component weights.
Add this regularizer to the view-agreement objective.

[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

§3 VICReg, SIGReg, VISReg and SWD

The method, the regularizer and the distance play different roles

Name	What it is	Distribution control	Scale treatment
VICReg	Full SSL objective	Variance floor + off-diagonal covariance	Explicit variance hinge
SIGReg	LeJEPA regularizer	Characteristic functions of random projections	Scale and shape in one Gaussian match
VISReg	Full SSL objective	Invariance + scale + shape + center	Separate weights and detached scale
SWD	Distance used by VISReg	Sorted 1D samples matched to Gaussian quantiles	Applied after feature normalization

Both SIGReg and VISReg go beyond second-order checks. Finite projections remain an approximation.

[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.
[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.

§3 Other mechanisms in the training recipe

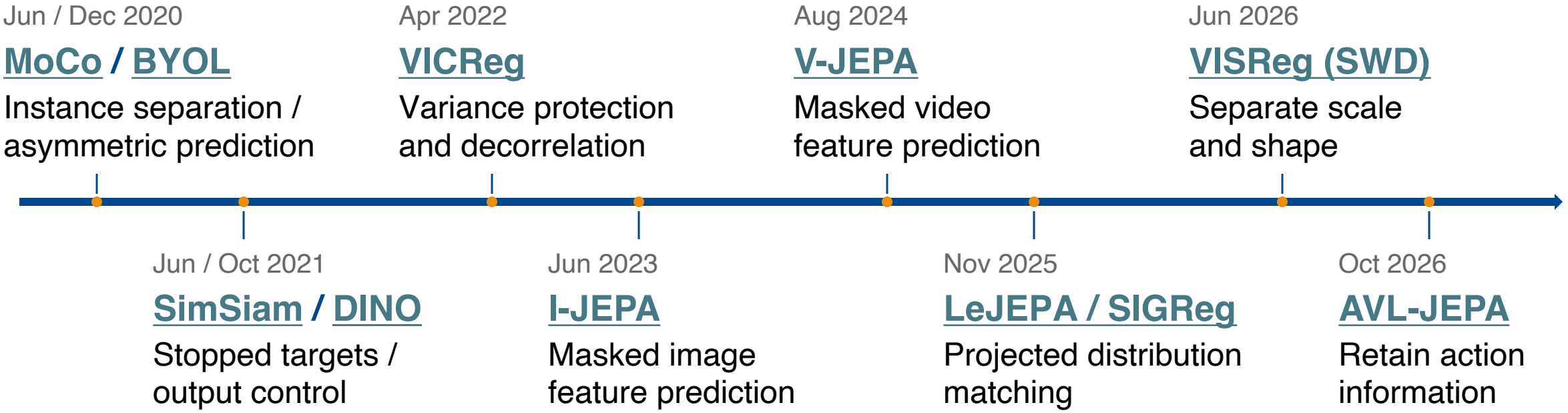
DINO shapes outputs; I-JEPA / V-JEPA shape the prediction task

Method	Target update	Additional mechanism	Main caution
BYOL	EMA; stop-gradient	Predictor + normalized regression	EMA alone fails in ablations
SimSiam	Shared weights; stop-gradient	Predictor + cosine matching	Removing stop-gradient collapses here
DINO	EMA; stop-gradient	Cross-view CE; center + sharpen	Entropy controls work jointly
I-JEPA / V-JEPA	EMA + stop-gradient	Predictor + masked feature task	Masks set the prediction content

[11] Bootstrap Your Own Latent - A New Approach to Self-Supervised Learning. December 2020. NeurIPS.
[13] Exploring Simple Siamese Representation Learning. June 2021. CVPR.
[7] Emerging Properties in Self-Supervised Vision Transformers. October 2021. ICCV.
[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[16] Revisiting Feature Prediction for Learning Visual Representations from Video. August 2024. TMLR.

§3 A chronology of collapse-control mechanisms

The arrow marks publication time, not a strict derivation chain



[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.

[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

§3 Preventing collapse: summary

Non-collapse is a constraint, not a task metric

Route	What it constrains	Advantage	Remaining limitation
Contrastive	Positive / negative gaps	Large queue (MoCo)	Pair-sampling assumptions
Asymmetric SSL	Gradients; target updates	No explicit negatives	Recipe-specific stability
VICReg	Std floor; covariance penalty	Symmetric, EMA-free training	Second-order control only
SIGReg	Gaussian tests on projections	No EMA / predictor in LeJEPA	Prior assumptions; finite slices
VISReg	Scale + normalized shape	Separate scale / shape weights	Sorting; task-specific weights

[12] Momentum Contrast for Unsupervised Visual Representation Learning. June 2020. CVPR.
[11] Bootstrap Your Own Latent - A New Approach to Self-Supervised Learning. December 2020. NeurIPS.
[13] Exploring Simple Siamese Representation Learning. June 2021. CVPR.
[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.
[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.

§4 Applications across modalities

4

4.1 Images and video

4.2 Language and robot planning

4.3 Factorization and action relevance

§4 The prediction interface changes

Inputs, targets and conditioning define the application

Application	Input	Prediction target	Extra signal
Image representation	Visible image patches	Masked patch features	Target positions
Video representation	Visible space-time patches	Hidden video features	Space-time positions
Vision-language	Image/video + query	Text embedding	Paired language
Robot planning	Visual state + action	Next-state features	Recorded actions + goals

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[16] Revisiting Feature Prediction for Learning Visual Representations from Video. August 2024. TMLR.
[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

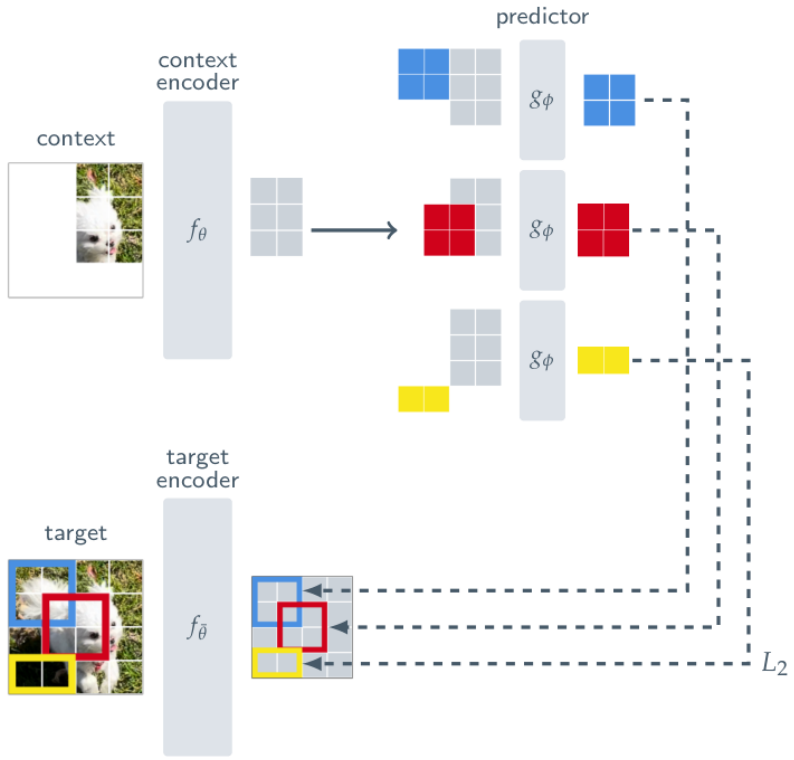
§4 I-JEPA: the image recipe

One context predicts multiple target regions

A context encoder sees visible patches.
A position-conditioned predictor fills target regions.
A full-image EMA encoder supplies the targets.

$$\mathcal{L}_{\text{paper}} = \frac{1}{M} \sum_{i=1}^{M} \sum_{j \in B_i} \| \hat{s}_j^{(i)} - \text{sg}(s_j) \|_2^2$$

M: target blocks. B_i: target patches. ŝ: prediction. s: target features.
sg: stop-gradient. The paper uses squared L2 matching.



Context encoder, predictor and target encoder, I-JEPA Fig. 3

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

§4 I-JEPA: masking shapes the task

Large target blocks encourage semantic prediction

Four target blocks cover 15–20% each.

The initial context covers 85–100%.

Remove target overlap from the visible context.



Original context and target examples, I-JEPA Fig. 4

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

§4 I-JEPA: frozen image representations

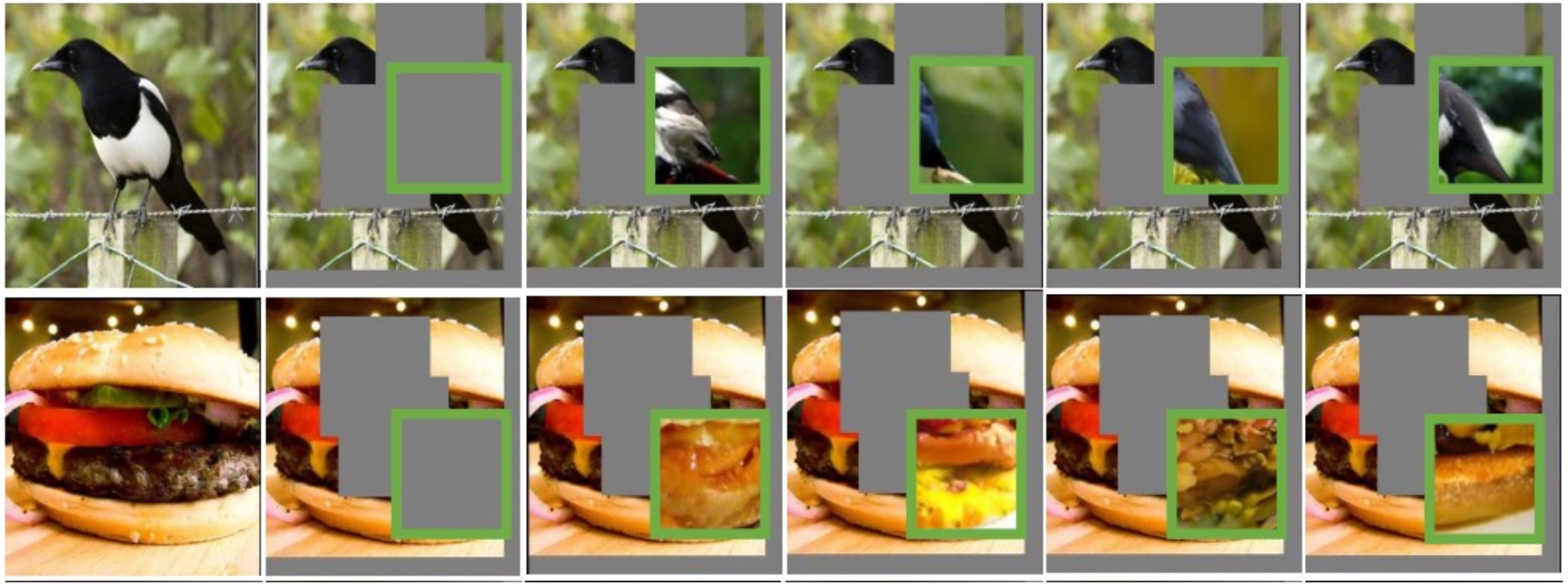
Strong linear probing with fewer pretraining epochs

Method	Backbone	Epochs	ImageNet top-1
MAE	ViT-L/16	1600	76.0%
data2vec	ViT-L/16	1600	77.3%
CAE	ViT-L/16	1600	78.1%
I-JEPA	ViT-L/16	600	77.5%

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

§4 I-JEPA: what the features retain

Predicted features can preserve semantic content



Two RCDM diagnostic examples, I-JEPA Fig. 6 detail

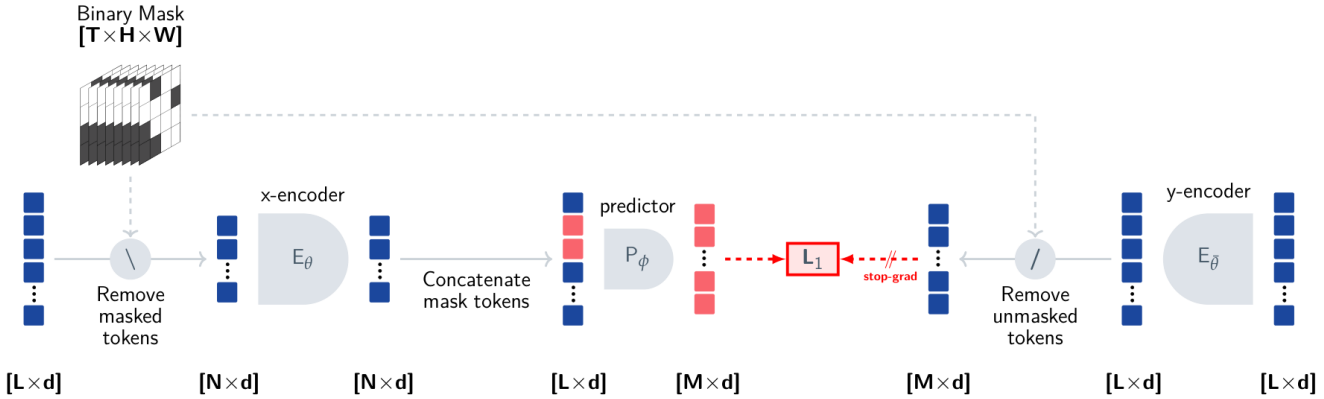
Samples retain object-level content while varying visual detail.

A separate diagnostic decoder produces these visualizations after pretraining.

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

§4 V-JEPA: prediction across space and time

Masked feature prediction extends to video



Spatiotemporal masked prediction, V-JEPA Fig. 3

$$\mathcal{L}_{\text{paper}} = \frac{1}{|B|} \sum_{j \in B} \| \hat{s}_j - \text{sg}(s_j) \|_1$$

Predict masked video features; the EMA teacher sees the full clip.
B: target tokens. ŝ: prediction. s: teacher features. sg: stop-gradient.
Use L1 matching and evaluate the frozen encoder with attentive probes.

[16] Revisiting Feature Prediction for Learning Visual Representations from Video. August 2024. TMLR.

§4 I-JEPA and V-JEPA

Video performance also depends on context and readout

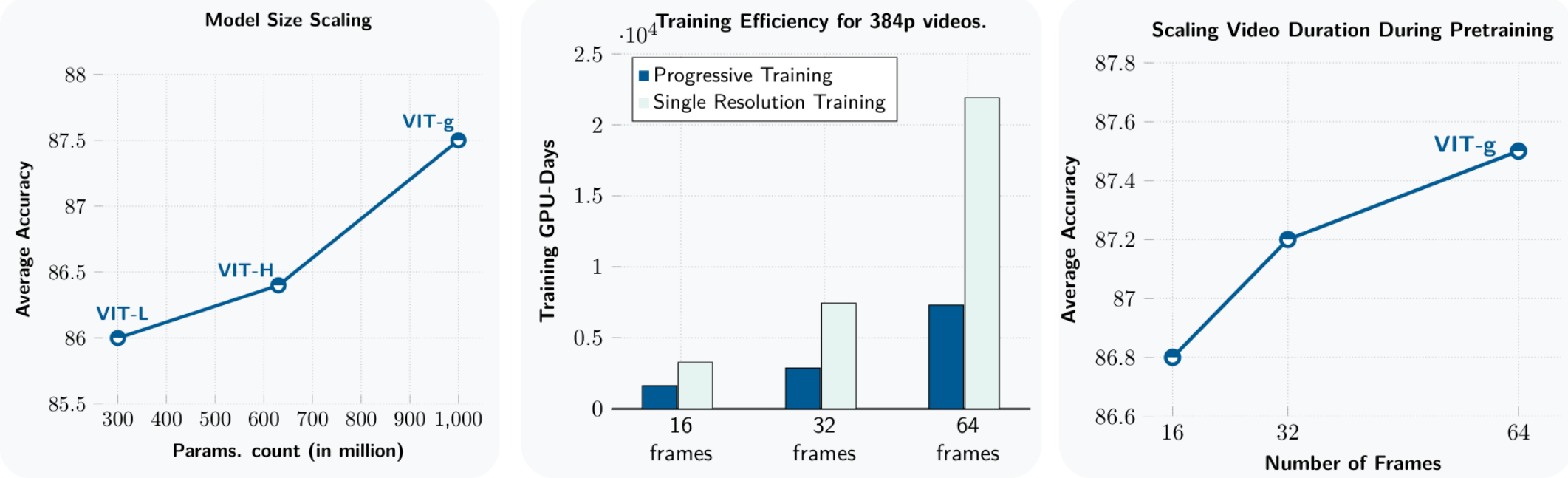
Aspect	I-JEPA	V-JEPA
Training input	A single image	A video clip
Masked targets	Spatial blocks	Spatiotemporal blocks
Feature loss	L2 regression	L1 regression
Target update	EMA	EMA
Core evaluation	Frozen linear probe	Frozen attentive probe

Different probes prevent a controlled comparison of image and video scores.

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[16] Revisiting Feature Prediction for Learning Visual Representations from Video. August 2024. TMLR.

§4 V-JEPA 2: scaling and video readouts

Scaling improves video readouts; actions require another stage



Scaling model, time and spatial resolution · V-JEPA 2, Fig. 5

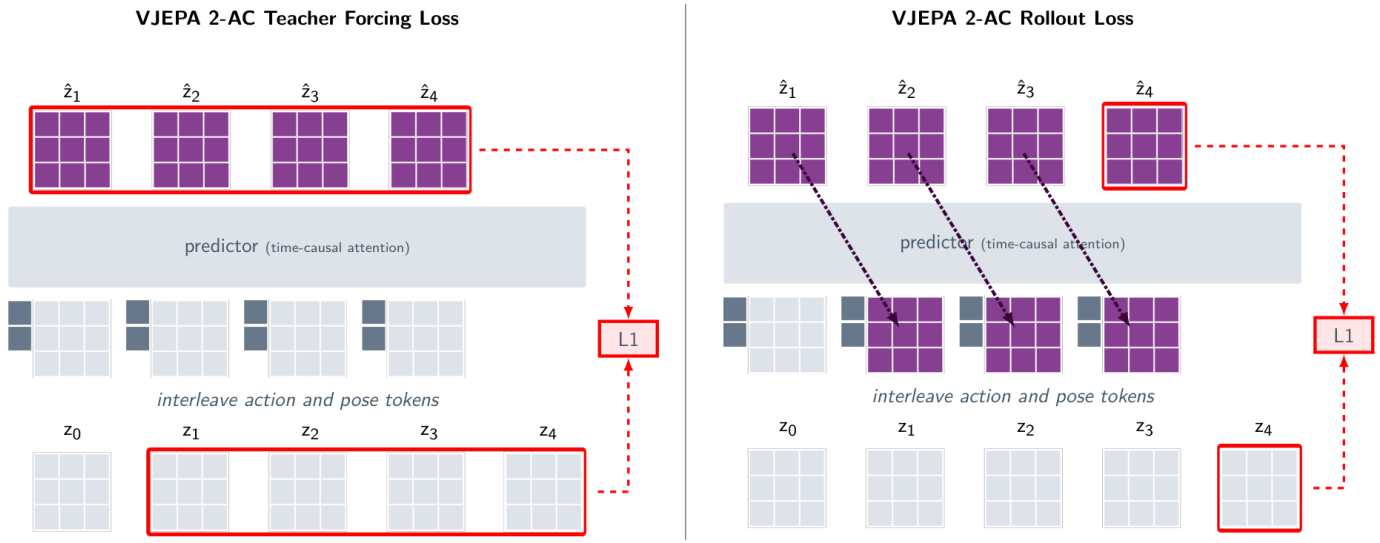
VideoMix22M, a ~1B encoder, and progressive spatiotemporal training.

SSv2: 77.3% top-1. EK100: 39.7% mean-class action recall@5.

[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

§4 V-JEPA 2-AC: learning action effects

Recorded actions turn visual features into dynamics



Teacher forcing (left) and rollout (right), V-JEPA 2 Fig. 6

Freeze the visual encoder; train an action-conditioned predictor.
Predict from visual features, recorded actions and robot state.

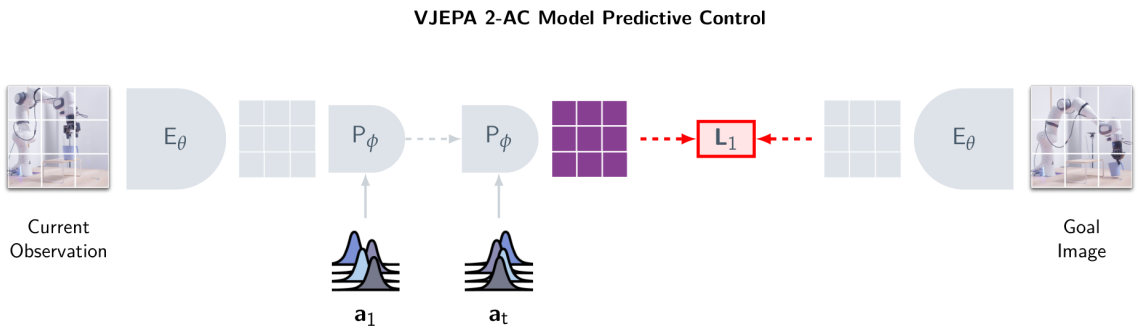
$$\mathcal{L} = \mathcal{L}_{\mathrm{TF}} + \mathcal{L}_{\mathrm{R}}$$

TF: observed history. R: predicted feedback.
Training uses 2 rollout steps (figure shows 4).

[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

§4 V-JEPA 2-AC: planning with a goal image

Search in latent space, then execute and observe



Goal-conditioned planning, Fig. 7



Real-robot rollouts, Fig. 10

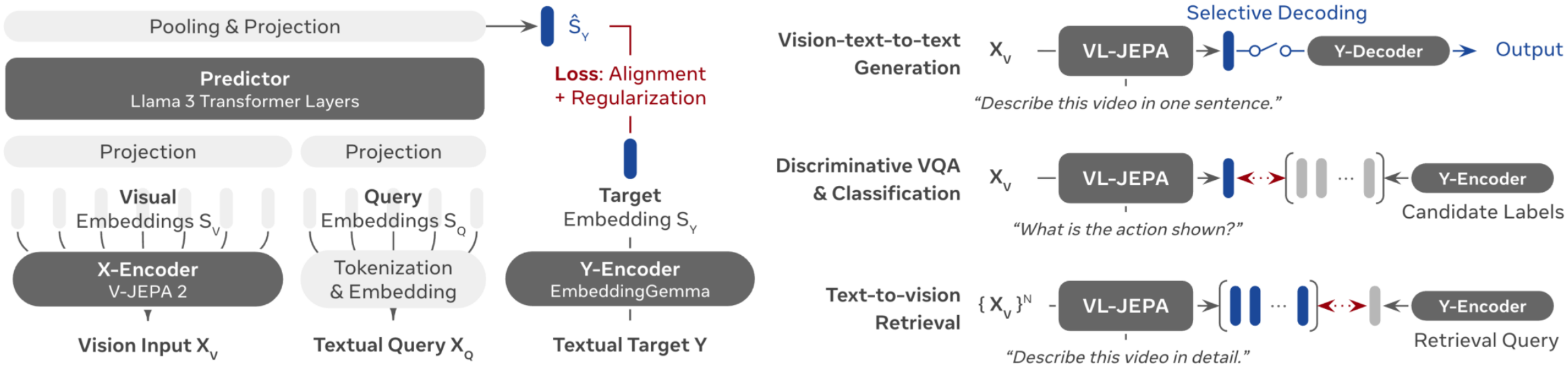
CEM searches action sequences using predicted distance to a visual goal.

Closed-loop execution replans after new observations.

[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

§4 VL-JEPA: predicting language embeddings

Predict meaning before decoding text



Architecture and downstream interfaces, VL-JEPA Fig. 2

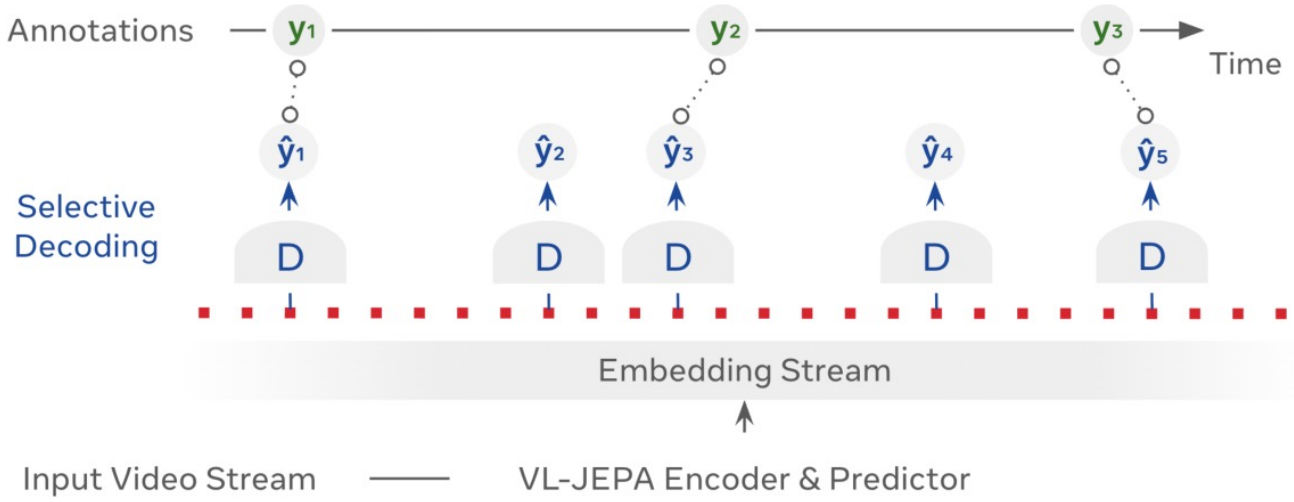
Visual features and a query predict a target text embedding.

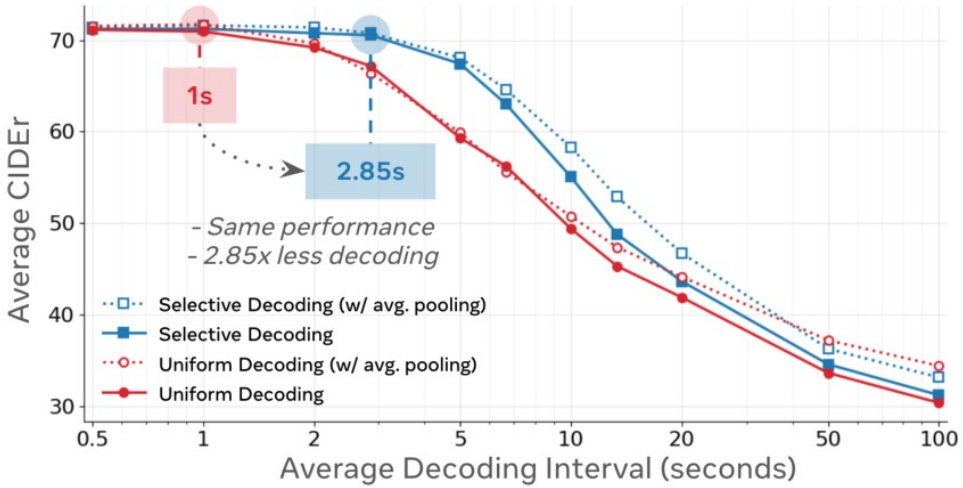
InfoNCE trains embedding prediction. A decoder renders text when needed.

[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.

§4 VL-JEPA: selective decoding

Update embeddings continuously, decode selectively





Selective decoding over video, VL-JEPA Fig. 4

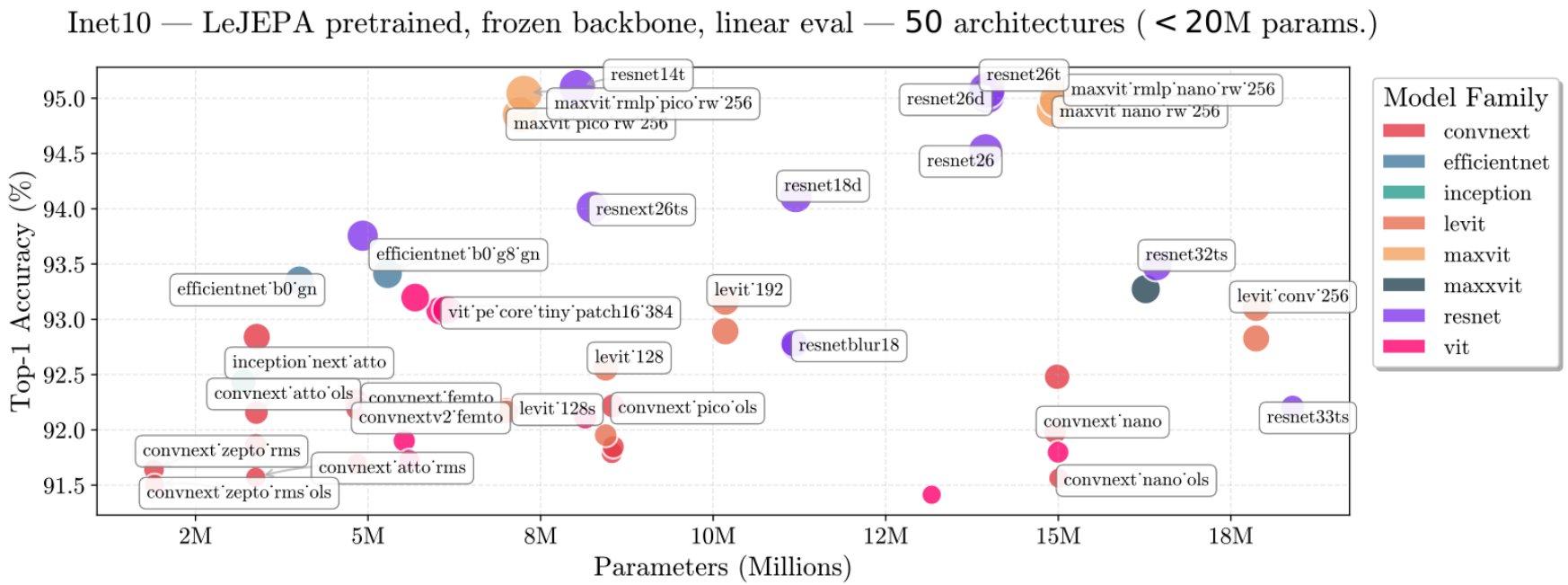
Temporal embedding structure selects representative decoding points.

The reported comparison uses about 2.85× fewer decoding calls at matched quality.

[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.

§4 LeJEPA: a compact practical recipe

Simple regularization can work across architectures



Frozen linear evaluation across architectures, LeJEPA Fig. 9

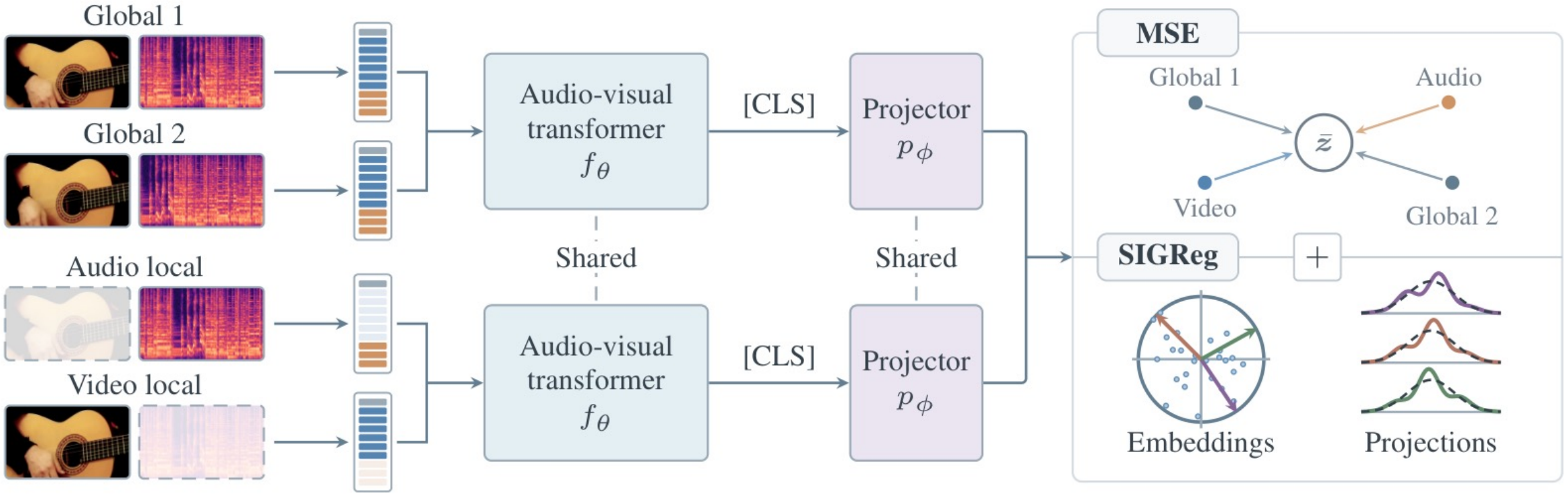
One prediction-versus-regularization tradeoff replaces several training heuristics.

Results span model families; optimizer settings still require selection.

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

§4 LeAVJEPA: audio-visual learning

A missing modality is another view of the same event



Match all four projected views to the mean of the two joint views.
Invariance aligns views of one clip; SIGReg keeps representations diverse across clips.

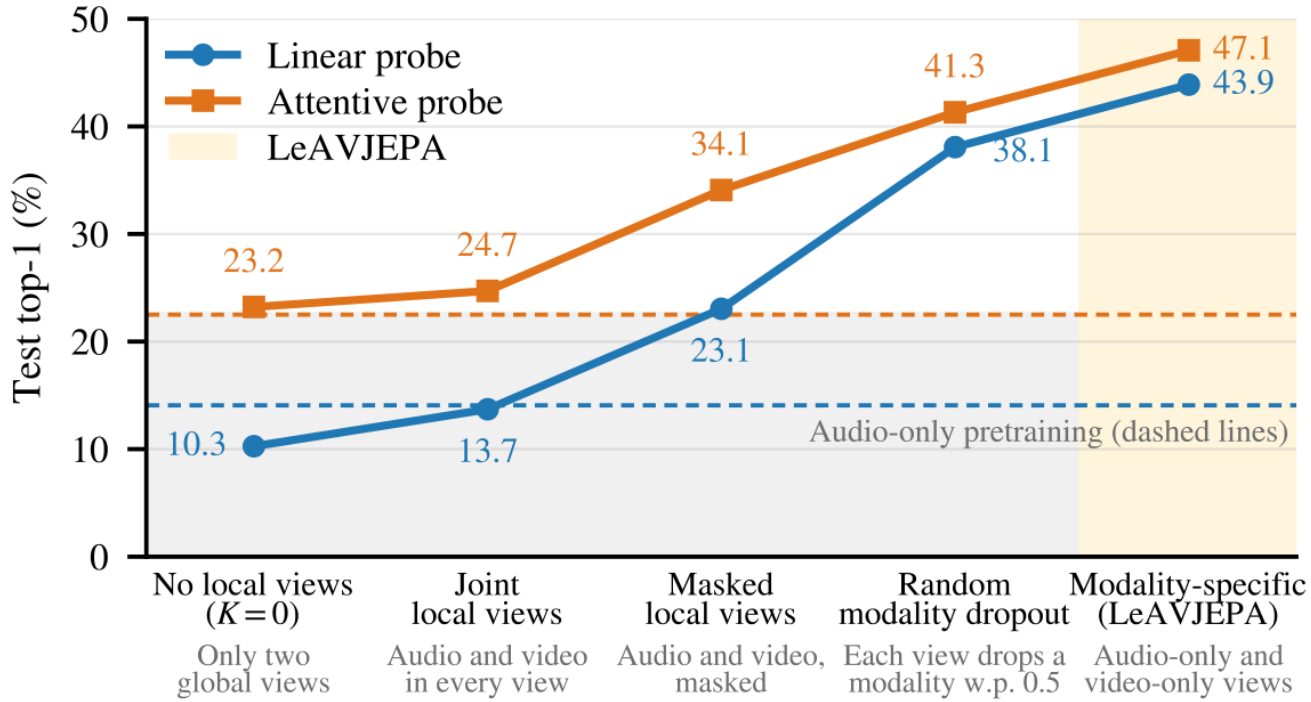
[32] LeAVJEPA: A Minimalist Architecture for Audio-Visual Self-Supervised Learning. October 2026. arXiv preprint.

§4 LeAVJEPA: evidence and limits

Modality-specific views improve cross-modal features

View ablation · VGGSound

ViT-B · online probes · epoch 30



Single-run ablation. Token counts and compute differ across view constructions.

Transfer · ViT-L

AudioSet-2M pretraining

Evaluation	Score
AudioSet-20K Frozen attentive · A-V	36.03 mAP
ESC-50 Frozen attentive · audio	91.3% acc.
VGGSound Fine-tuned · A-V	61.1% top-1

[32] LeAVJEPA: A Minimalist Architecture for Audio-Visual Self-Supervised Learning. October 2026. arXiv preprint.

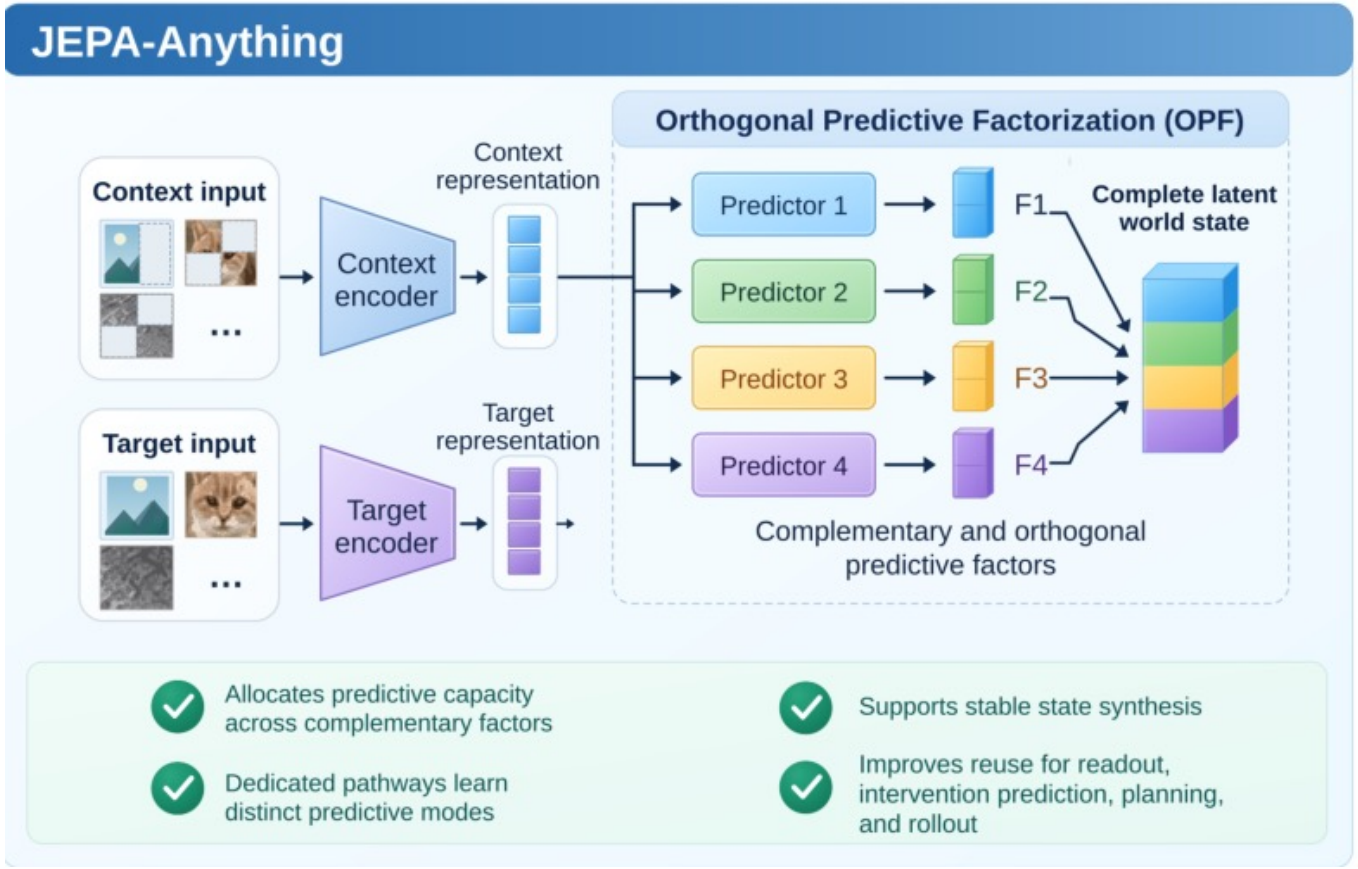
§4 JEPA-Anything: predictive factorization

Add factor structure to each domain's prediction task

Split targets into learned subspaces; predict each with a separate head.

Add orthogonality and activity penalties to each domain's base loss.

Train separate instances across seven domains.

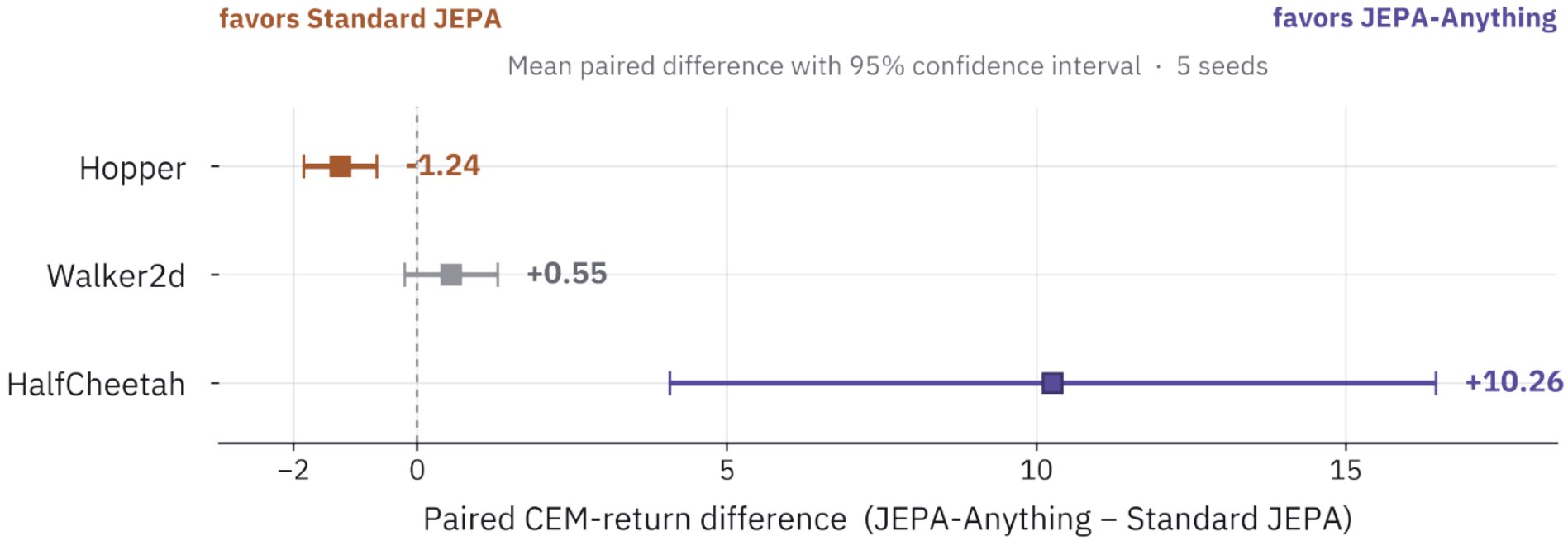


Orthogonal Predictive Factorization, Fig. 1 detail

[18] JEPA-Anything: Learning Predictive Models across Different Worlds. September 2026. arXiv preprint.

§4 JEPA-Anything: prediction and decisions

Predictive gains do not guarantee better decisions



Paired return differences under matched CEM · JEPA-Anything, Fig. 6

Matched CEM control gives task-dependent gains.

Hopper worsens; Walker2d's 95% interval crosses zero.

[18] JEPA-Anything: Learning Predictive Models across Different Worlds. September 2026. arXiv preprint.

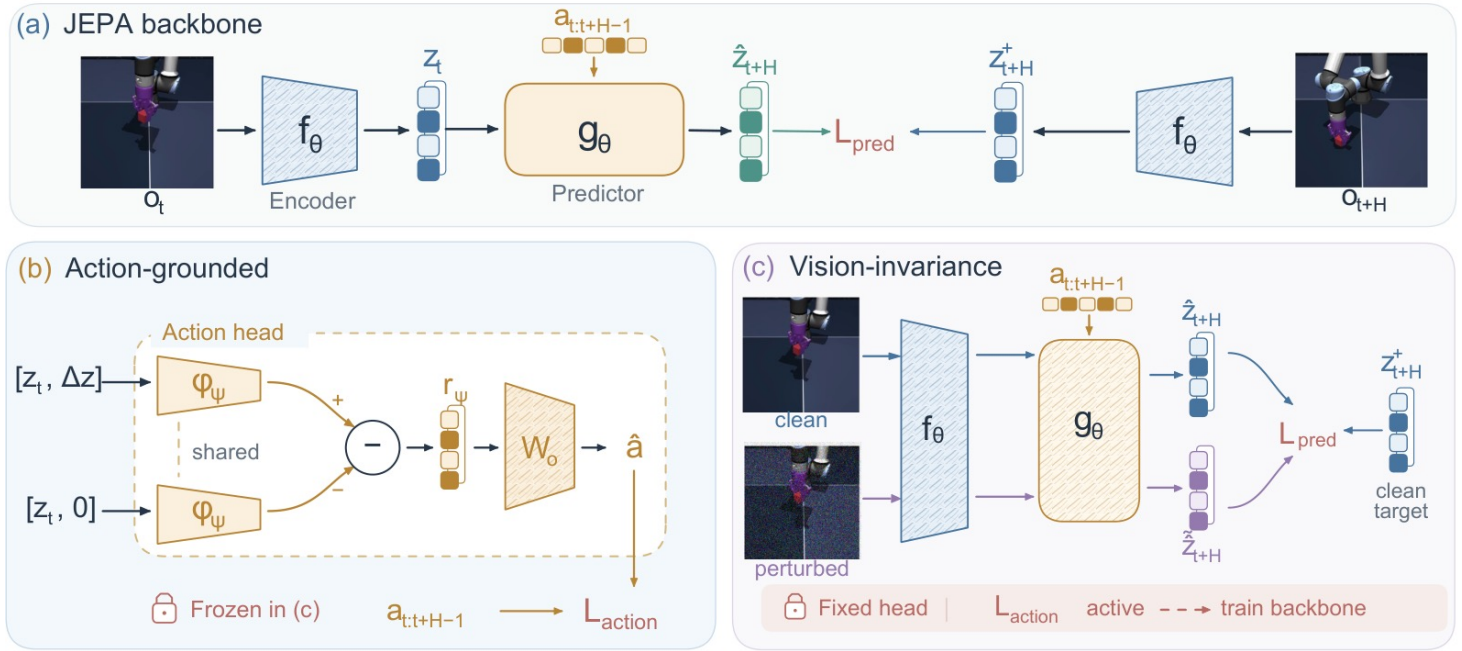
§4 AVL-JEPA: action-relevant information

Preserve action effects while suppressing visual nuisance

Recover actions from latent transitions, subtracting a state-only baseline.

Freeze the action head during noisy-input alignment; keep its loss active.

Planning retains the LEWM interface; evaluation is in simulation.

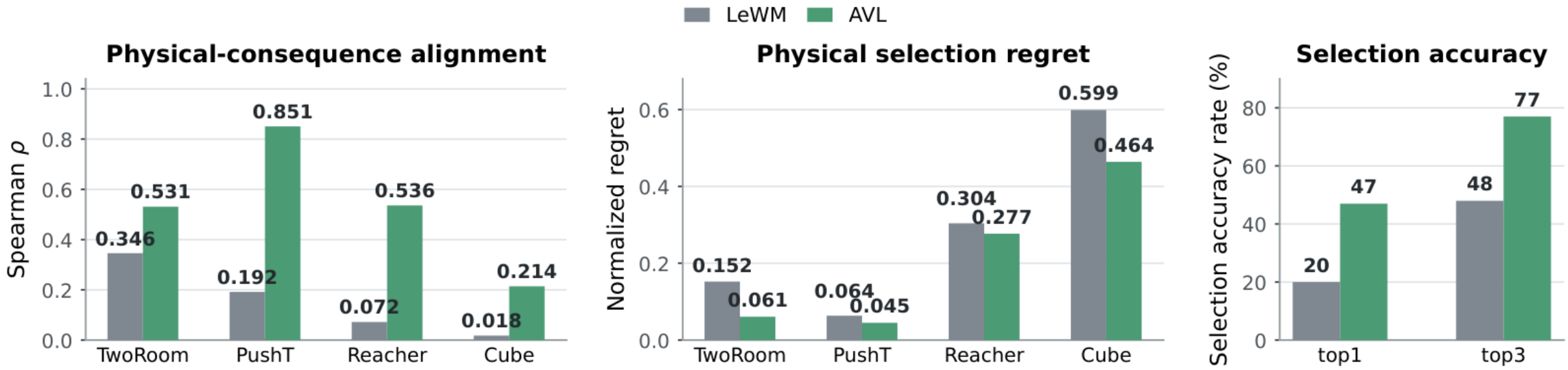


Action grounding and visual invariance, AVL-JEPA Fig. 2

[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

§4 AVL-JEPA: test action consequences

Useful dynamics need intervention-based tests



Matched action consequences in simulation · AVL-JEPA, Fig. 4

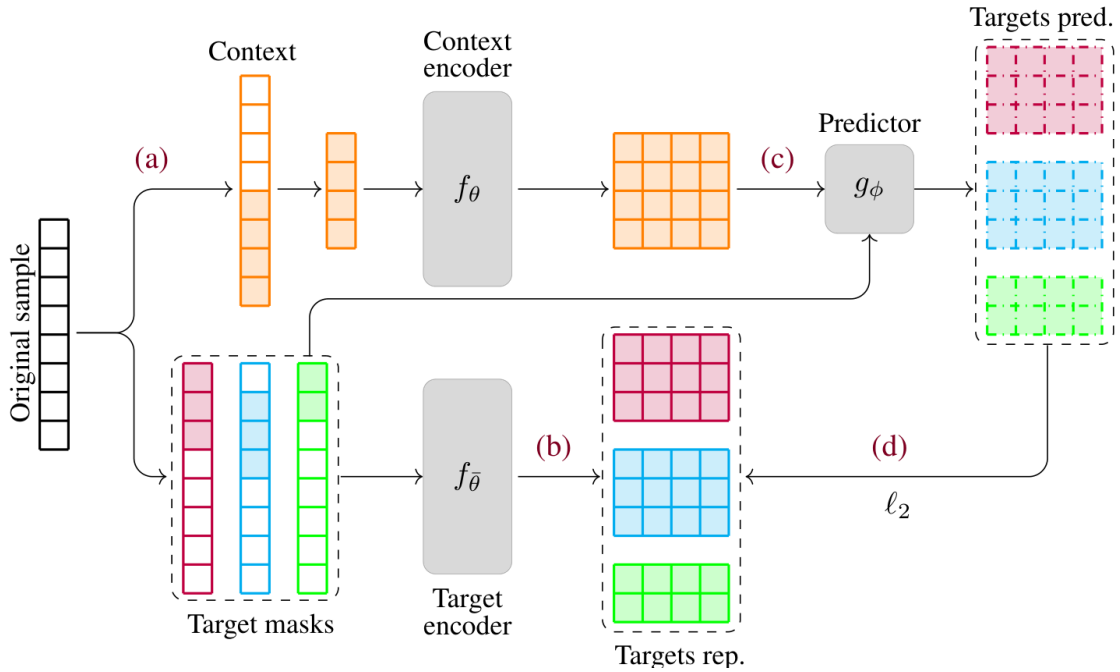
Hold the state fixed; rank shared actions against simulator outcomes.

Separate nuisance robustness from clean-task success.

[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

§4 T-JEPA: learning from table columns

Masked feature prediction avoids designing synthetic views



Feature masking and latent prediction · T-JEPA, Fig. 1

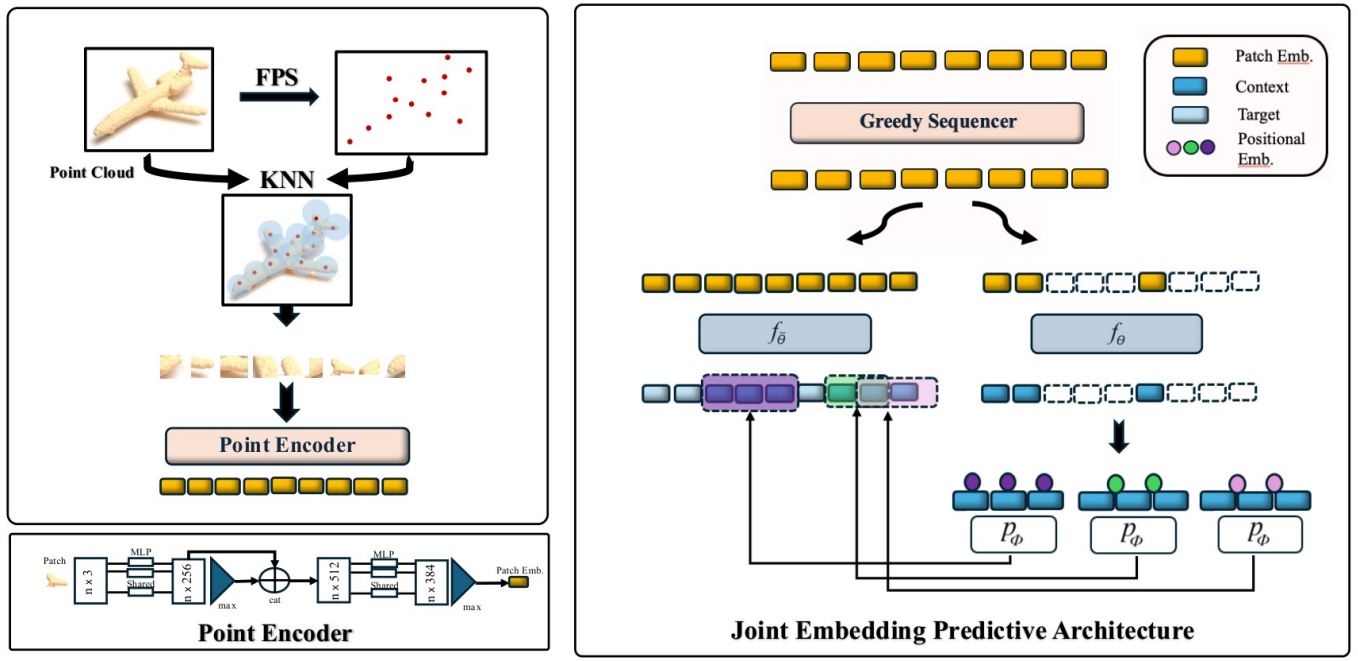
Predict held-out column embeddings with an EMA target.

Most neural readouts improve; Higgs exposes negative transfer.

[28] T-JEPA: Augmentation-Free Self-Supervised Learning for Tabular Data. April 2025. ICLR.

§4 Point-JEPA: learning from 3D point clouds

Spatial ordering makes masked 3D prediction practical



Point encoding, sequencing and JEPA · Point-JEPA, Fig. 2

A greedy sequencer builds contiguous context and target blocks.

Strong object classification; part segmentation remains weaker.

[29] Point-JEPA: A Joint Embedding Predictive Architecture for Self-Supervised Learning on Point Cloud. February 2025. WACV.

§4 Applications: learned representations

Judge the trained interface and the evidence it supports

Work	Target and structure	Main evidence	Key limitation	Contribution to the field
I-JEPA	Block targets; EMA	Frozen probes	No dynamics	Images: no pixel decoder
V-JEPA	Space-time features	Frozen attentive probes	No action conditioning	Video features without frame synthesis
V-JEPA 2	Data/model scaling	SSv2 / EK100 readouts	Robot data for actions	Video backbone at scale
LeJEPA	Prediction + SIGReg	Cross-architecture probes	Conditional theory	Image SSL without EMA or stop-gradient
T-JEPA	Column targets	Neural readouts	Higgs mixed	Tables: no view design
Point-JEPA	Point ordering	Classification	Weaker mIoU	Unordered 3D features

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[16] Revisiting Feature Prediction for Learning Visual Representations from Video. August 2024. TMLR.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.
[28] T-JEPA: Augmentation-Free Self-Supervised Learning for Tabular Data. April 2025. ICLR.
[29] Point-JEPA: A Joint Embedding Predictive Architecture for Self-Supervised Learning on Point Cloud. February 2025. WACV.

§4 Applications: interfaces and decisions

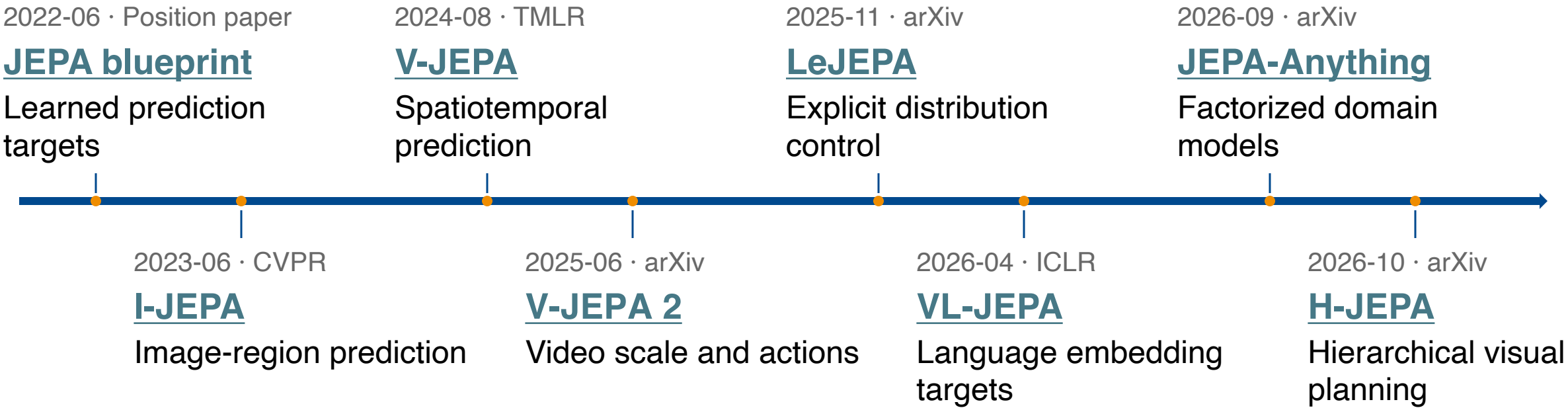
Additional structure creates specific usable capabilities

Work	Additional structure	Resulting use	Evidence boundary	Contribution to the field
V-JEPA 2-AC	Frozen encoder + action predictor	Goal-image CEM on Franka	Short horizons; image goals	Robotics: video features for action search.
VL-JEPA	Text-embedding InfoNCE	Retrieval, VQA, captioning	Offline decoding counts	VLMs: semantic updates with selective text decoding.
JEPA-Anything	Subspaces + factor heads	Intervention prediction; CEM	Separate models; mixed gains	Cross-domain: a shared factor interface.
AVL-JEPA	Action anchor + noisy alignment	Simulated MPC under visual shifts	Simulation only	Control: ranks action consequences.

[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.
[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.
[18] JEPA-Anything: Learning Predictive Models across Different Worlds. September 2026. arXiv preprint.
[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

§4 JEPA literature timeline

Selected publications, targets and constraints



[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[16] Revisiting Feature Prediction for Learning Visual Representations from Video. August 2024. TMLR.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.
[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.
[18] JEPA-Anything: Learning Predictive Models across Different Worlds. September 2026. arXiv preprint.
[8] H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning. October 2026. arXiv preprint.

§5 Synthesis and open questions

5

5.1 What the literature establishes

↓

5.2 What the representation must retain

↓

5.3 Questions for co-design

§5 Three choices shape representation learning

Architecture alone does not specify the learned representation

Choice	Representative alternatives	What it changes
Prediction target	Pixels, teacher features, text embeddings	What the model must explain
Constraint	Negatives, asymmetry, moments, distributions	Which degenerate solutions are removed
Task interface	Probe, decoder, action model, planner	How the representation becomes useful

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.
[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.
[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

§5 What non-collapse does not establish

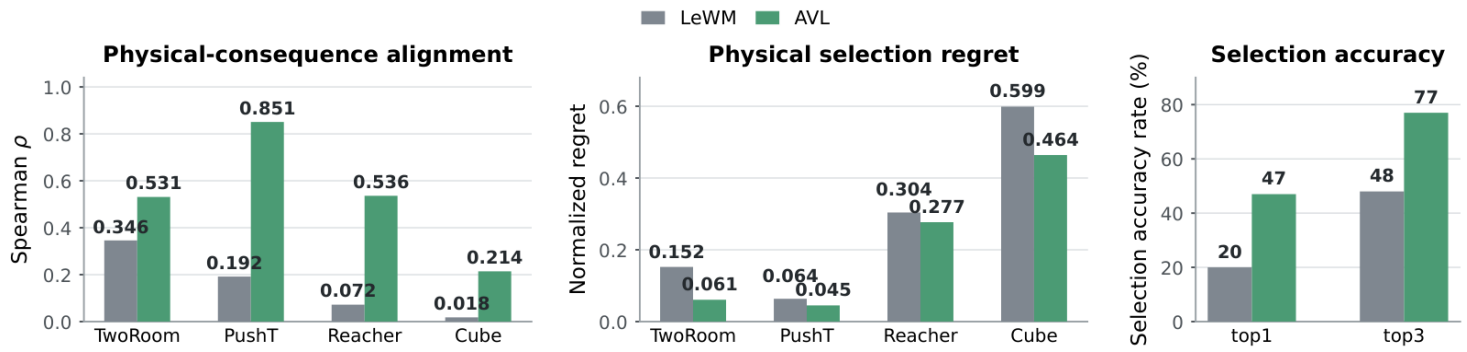
Information quality needs task-specific evidence

Variance and rank describe spread and dimensionality.

Dense tasks require spatial detail.

Control requires action-sensitive state changes.

Transfer tests whether those properties survive a new setting.



Physical-consequence alignment diagnostics, AVL-JEPA Fig. 4

[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.

[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

§5 Representation–generation co-design

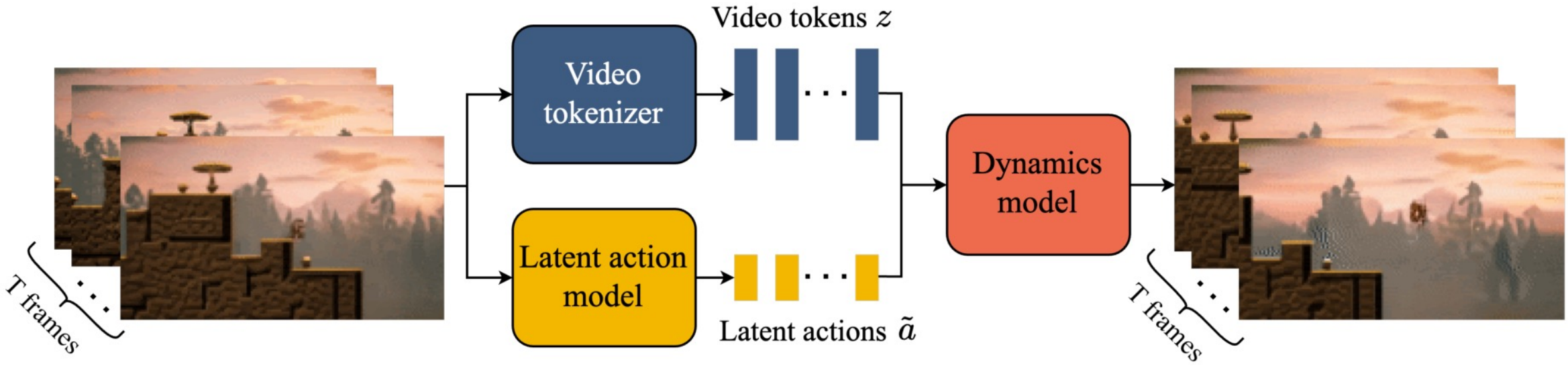
A useful latent must satisfy more than one objective

Desired property	A relevant lesson	What to test
Preserve information	Predictive targets can discard detail	Reconstruction + semantic and spatial probes
Make learning tractable	Scale and shape affect optimization	Gradient scale + rank + task loss
Support the final task	Actions and readouts define utility	Generation quality + sampling behavior

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.
[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.
[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

§5 GENIE: Interactive Worlds from Video

Video prediction becomes an interactive environment



Video tokenizer, latent action model and dynamics model, GENIE Fig. 3

Learn discrete latent actions from unlabeled video.

Predict video tokens; decode action-controlled futures.

Scale to an 11B model for image-prompted 2D worlds.

[20] Genie: Generative Interactive Environments. July 2024. ICML.

§5 World Models toward Embodied Intelligence

Connect perception, interactive simulation and physical-world action

	V-JEPA 2-AC	GENIE
Prediction target	Visual features	Video tokens → frames
Control interface	Robot actions + planning	Learned latent action codes
Reported evidence	Specified real-robot tasks	Interactive game / robot video

Embodied intelligence requires grounded actions and evaluation in the physical world.

[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.
[20] Genie: Generative Interactive Environments. July 2024. ICML.

§5 Takeaways

The central question is which information remains usable

JEPA predicts learned representations within a broader intelligent system.

Stable training needs explicit constraints or a suitable asymmetric recipe.

Images, video, language and control require different targets and interfaces.

Useful representations must preserve the information required by the task.

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.
[15] VISReg: Variance-Invariance-Sketching Regularization for JEPA training. June 2026. arXiv preprint.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.
[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.
[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

Personal Reflections

Non-collapse does not establish task sufficiency

Design risk: diverse features may still discard action-relevant changes.

Proposed test: vary mask or horizon, then probe action effects at matched latent statistics.

[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.
[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.

Personal Reflections

What does the input–target correspondence contribute?

Shared-checkpoint test: permute target samples while preserving batch statistics.

Long-run control: use one frozen teacher and compare transfer at equal compute.

[5] data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language. July 2022. ICML.
[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.
[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

Personal Reflections

Do action rankings transfer to goal-directed control?

Existing evidence: AVL-JEPA tests physical-response rankings.

Open test: do goal-cost rankings track closed-loop gains across prediction horizons?



[19] AVL-JEPA: Preventing Causal Dynamics Information Collapse In Joint Embedding Predictive Architecture World Models. October 2026. arXiv preprint.
[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.
[18] JEPA-Anything: Learning Predictive Models across Different Worlds. September 2026. arXiv preprint.

References: selected contributors

Follow the people behind each design choice

Research line	Teams in the cited papers	Selected contributors
Images & video	Meta FAIR	Mahmoud Assran · Adrien Bardes · Nicolas Ballas
Theory & regularization	Brown · NYU · Meta FAIR	Randall Balestriero · Yann LeCun
Vision-language	Meta FAIR · HKUST · Sorbonne	Delong Chen · Mustafa Shukor · Pascale Fung
Tabular learning	Paris-Saclay · CentraleSupélec · LISN	Hugo Thimonier · José Lucas De Melo Costa · Fabrice Popineau
3D point clouds	Saint Mary's University	Ayumu Saito · Prachi Kudeshia · Jiju Poovvancheri

[4] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. June 2025. arXiv preprint.

[14] LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics. November 2025. arXiv preprint.

[17] VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. April 2026. ICLR.

[28] T-JEPA: Augmentation-Free Self-Supervised Learning for Tabular Data. April 2025. ICLR.

[29] Point-JEPA: A Joint Embedding Predictive Architecture for Self-Supervised Learning on Point Cloud. February 2025. WACV.

References: reading and viewing

Use diagrams for orientation, papers for evidence

Entry point	What to inspect
I-JEPA · Meta, June 2023	Inspect the masking diagram and semantic-feature visualizations.
V-JEPA · Meta, February 2024	Trace masked video training to frozen attentive evaluation.
V-JEPA 2 · Meta, June 2025	Distinguish video pretraining, action post-training and robot demos.
Yann LeCun · Brown, April 2026	Author lecture on world models; official recording entry.

[21] I-JEPA: The first AI model based on Yann LeCun's vision for more human-like AI. June 2023. Meta AI Blog.

[31] V-JEPA: The next step toward Yann LeCun's vision of advanced machine intelligence (AMI). February 2024. Meta AI Blog.

[22] Introducing the V-JEPA 2 world model and new benchmarks for physical reasoning. June 2025. Meta AI Blog.

[30] World Models: Enabling the AI Revolution. April 2026. Brown University · Lemley Lecture.

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[1] A Path Towards Autonomous Machine Intelligence. June 2022. Position paper.

[2] Introduction to latent variable energy-based models: a path toward autonomous machine intelligence. October 2024. J. Stat. Mech.

[3] Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. June 2023. CVPR.

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[5] data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language. July 2022. ICML.

[6] Masked Autoencoders Are Scalable Vision Learners. June 2022. CVPR.

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[7] Emerging Properties in Self-Supervised Vision Transformers. October 2021. ICCV.

[8] H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning. October 2026. arXiv preprint.

[9] You Don't Need Strong Assumptions: Visual Representation Learning via Temporal Differences. June 2026. arXiv preprint.

[10] VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. April 2022. ICLR.

[11] Bootstrap Your Own Latent - A New Approach to Self-Supervised Learning. December 2020. NeurIPS.

[12] Momentum Contrast for Unsupervised Visual Representation Learning. June 2020. CVPR.

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[13] Exploring Simple Siamese Representation Learning. June 2021. CVPR.

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[24] An Introduction to the JEPA Series. February 2026. Bilibili.

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[27] The Information Geometry of Representations in JEPA, Energy-Based Models, Diffusion Models, Transformers and VAEs [Chinese Subtitles]. July 2026. Bilibili.

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